

Quantum Field Theory – Homework 1

Name: _____ Due Date: 2025.10.27

The bullet points under each problem are structured hints or guiding points to help you know what to include in your solution.

Problem 1. Classical Field Theories and the Principle of Locality (20 points)

Summarize the key concepts presented in **Lecture 1** on *The Principle of Locality and Fields* and **Lecture 2** on *Classical Field Theories*. Your summary should demonstrate a clear understanding of how these ideas motivate the formulation of modern field theories.

- Discuss the motivation for introducing field theory starting from classical mechanics.
- Explain the **principle of locality** and how it constrains the form of physical interactions and Lagrangians.
- Provide at least one example (e.g., Maxwells equations or Einsteins gravity) that illustrates how the principle of locality appears in physical laws.
- Describe briefly the general form of the action and Lagrangian density in classical field theory, and explain why only first-order time derivatives are usually included.

Problem 2. Free-Fall Motion in a Gravitational Field (20 points)

Consider a particle of mass m moving vertically under a uniform gravitational field g . The action is

$$S[z] = \int_{t_1}^{t_2} L(z, \dot{z}, t) dt, \quad L(z, \dot{z}, t) = \frac{1}{2}m\dot{z}^2 - mgz.$$

1. Find the conjugate momentum p and write down the Hamiltonian $H(z, p)$ for this system.

Solution: The conjugate momentum is

$$p \equiv \frac{\partial L}{\partial \dot{z}} = m\dot{z}.$$

The Hamiltonian is

$$H(z, p) \equiv p\dot{z} - L = p \left(\frac{p}{m} \right) - \left[\frac{1}{2}m \left(\frac{p}{m} \right)^2 - mgz \right] = \frac{p^2}{2m} + mgz.$$

2. Using the Euler-Lagrange equation, derive the equation of motion for $z(t)$.

Solution: Euler-Lagrange equation:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{z}} \right) - \frac{\partial L}{\partial z} = 0 \quad \Rightarrow \quad \frac{d}{dt}(m\dot{z}) - (-mg) = 0 \quad \Rightarrow \quad m\ddot{z} + mg = 0.$$

Hence the equation of motion is

$$\ddot{z} = -g.$$

3. Solve the equation of motion and express $z(t)$ in terms of the initial position $z(0)$ and initial velocity $\dot{z}(0)$.

Solution: Integrate twice and impose the initial data $z(0)$ and $\dot{z}(0)$:

$$\dot{z}(t) = \dot{z}(0) - gt, \quad z(t) = z(0) + \dot{z}(0)t - \frac{1}{2}gt^2,$$

where $z(t)$ is the solution of the equation of motion, which also describes the trajectory of the particle.

Problem 3. Deriving Maxwells Equations from the Electromagnetic Action (20 points)

Consider the electromagnetic field in vacuum, described by the four-potential $A_\mu(x)$ and the field-strength tensor

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

The action of the electromagnetic field (in natural units $c = \hbar = 1$) is

$$S[A] = \int d^4x \mathcal{L}, \quad \mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}.$$

1. By varying the action with respect to A_ν , derive the Euler-Lagrange equation

$$\partial_\mu F^{\mu\nu} = 0,$$

which corresponds to the **inhomogeneous Maxwell equations** in vacuum.

Solution: We vary the action with respect to A_ν and assume δA_ν vanishes on the boundary:

$$S[A] = \int d^4x \mathcal{L}, \quad \mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

Compute δS :

$$\delta S = \int d^4x \left(-\frac{1}{4} \delta(F_{\mu\nu} F^{\mu\nu}) \right) = \int d^4x \left(-\frac{1}{2} F^{\mu\nu} \delta F_{\mu\nu} \right),$$

using

$$\begin{aligned} \delta(F_{\mu\nu} F^{\mu\nu}) &= \delta F^{\mu\nu} F_{\mu\nu} + F^{\mu\nu} \delta F_{\mu\nu} = \delta F^{\mu\nu} F_{\mu\nu} + g^{\mu\rho} g^{\nu\sigma} F_{\mu\nu} \delta F_{\rho\sigma} \\ &= \delta F^{\mu\nu} F_{\mu\nu} + F^{\rho\sigma} \delta F_{\rho\sigma} = 2 F^{\mu\nu} \delta F_{\mu\nu}. \end{aligned}$$

Now

$$\delta F_{\mu\nu} = \partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu,$$

so

$$\delta S = -\frac{1}{2} \int d^4x F^{\mu\nu} (\partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu) = - \int d^4x F^{\mu\nu} \partial_\mu \delta A_\nu,$$

where antisymmetry $F^{\mu\nu} = -F^{\nu\mu}$ makes the two terms equal and gives the factor of

2. Integrate by parts and drop the boundary term:

$$\delta S = \int d^4x (\partial_\mu F^{\mu\nu}) \delta A_\nu.$$

Since δA_ν is arbitrary in the bulk, stationarity $\delta S = 0$ implies

$$\partial_\mu F^{\mu\nu} = 0.$$

2. The tensor $F_{\mu\nu}$ is antisymmetric by definition:

$$F_{\mu\nu} = -F_{\nu\mu}.$$

Using this property, and assuming that the partial derivatives commute ($\partial_\alpha\partial_\beta = \partial_\beta\partial_\alpha$), show that $F_{\mu\nu}$ satisfies

$$\partial_\alpha F_{\beta\gamma} + \partial_\beta F_{\gamma\alpha} + \partial_\gamma F_{\alpha\beta} = 0.$$

This identity represents the **homogeneous Maxwell equations**.

Solution: By definition,

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad \Rightarrow \quad F_{\mu\nu} = -F_{\nu\mu}.$$

Compute the cyclic sum:

$$\partial_\alpha F_{\beta\gamma} + \partial_\beta F_{\gamma\alpha} + \partial_\gamma F_{\alpha\beta} = \partial_\alpha(\partial_\beta A_\gamma - \partial_\gamma A_\beta) + \partial_\beta(\partial_\gamma A_\alpha - \partial_\alpha A_\gamma) + \partial_\gamma(\partial_\alpha A_\beta - \partial_\beta A_\alpha).$$

Group like terms:

$$(\partial_\alpha\partial_\beta A_\gamma - \partial_\beta\partial_\alpha A_\gamma) + (\partial_\beta\partial_\gamma A_\alpha - \partial_\gamma\partial_\beta A_\alpha) + (\partial_\gamma\partial_\alpha A_\beta - \partial_\alpha\partial_\gamma A_\beta).$$

Assuming partial derivatives commute, $\partial_\mu\partial_\nu = \partial_\nu\partial_\mu$, each parenthesis vanishes. Hence

$$\partial_\alpha F_{\beta\gamma} + \partial_\beta F_{\gamma\alpha} + \partial_\gamma F_{\alpha\beta} = 0$$

which is the homogeneous Maxwell identity (Bianchi identity).

3. Using the identifications

$$E_i = F_{i0}, \quad B^i = \frac{1}{2}\varepsilon^{ijk}F_{jk},$$

express the four-dimensional equations from parts (1) and (2) in 3+1 form, and verify

that they reproduce the familiar vacuum Maxwell equations:

$$\nabla \cdot \mathbf{E} = 0, \quad \nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0, \quad \nabla \times \mathbf{B} - \partial_t \mathbf{E} = 0.$$

Solution: From part (1) we have the inhomogeneous Maxwell equations in vacuum:

$$\partial_\mu F^{\mu\nu} = 0.$$

Split by components. the covariant components of $F_{\mu\nu}$ is

$$F_{\mu\nu} = \begin{pmatrix} 0 & E_x & E_y & E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{pmatrix}.$$

Gauss for $\mathbf{E}$: set $\nu = 0$:

$$0 = \partial_\mu F^{\mu 0} = \partial_i F^{i0} = \partial_i (-E_i) \Rightarrow \nabla \cdot \mathbf{E} = 0.$$

Ampère - Maxwell (no sources): set $\nu = j$:

$$0 = \partial_\mu F^{\mu j} = \partial_0 F^{0j} + \partial_i F^{ij} = \partial_t E_j + \partial_i (-\varepsilon^{ijk} B_k) = \partial_t E_j - (\nabla \times \mathbf{B})_j,$$

so

$$\nabla \times \mathbf{B} - \partial_t \mathbf{E} = 0.$$

From part (2) we have the homogeneous identity (Bianchi identity)

$$\partial_\alpha F_{\beta\gamma} + \partial_\beta F_{\gamma\alpha} + \partial_\gamma F_{\alpha\beta} = 0.$$

Again split by components.

Gauss for $\mathbf{B}$: take purely spatial indices $(\alpha, \beta, \gamma) = (i, j, k)$:

$$0 = \partial_i F_{jk} + \partial_j F_{ki} + \partial_k F_{ij} = \partial_i (\varepsilon_{jkl} B^\ell) + \text{cyc} = \varepsilon_{jkl} \partial_i B^\ell + \text{cyc} \Rightarrow \nabla \cdot \mathbf{B} = 0.$$

(Contract with ε^{ink} and use $\varepsilon^{ijk}\varepsilon_{jkl} = 2\delta^i_\ell$.)

Faradays law: take one time and two space indices, e.g. $(\alpha, \beta, \gamma) = (0, i, j)$:

$$0 = \partial_0 F_{ij} + \partial_i F_{j0} + \partial_j F_{0i} = \partial_t (\varepsilon_{ijk} B^k) - \partial_i E_j + \partial_j E_i.$$

Contract with $\frac{1}{2}\varepsilon^{\ell ij}$ to get

$$(\partial_t \mathbf{B})^\ell + (\nabla \times \mathbf{E})^\ell = 0 \quad \Rightarrow \quad \nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0.$$

Collecting results:

$$\boxed{\nabla \cdot \mathbf{E} = 0, \quad \nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0, \quad \nabla \times \mathbf{B} - \partial_t \mathbf{E} = 0}$$

which are the familiar vacuum Maxwell equations in $3 + 1$ form.

4. Briefly explain why the action $S[A]$ represents a **local and Lorentz-invariant** field theory.

Solution: The action

$$S[A] = \int d^4x \mathcal{L}, \quad \mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu},$$

is both **local** and **Lorentz invariant** because:

- *Locality:* The Lagrangian density $\mathcal{L}$ depends only on the field $A_\mu(x)$ and its derivatives $\partial_\mu A_\nu(x)$ evaluated at the same spacetime point x . There are no nonlocal terms such as integrals or functions of fields at different points $x' \neq x$. Hence, interactions are local.
- *Lorentz invariance:* The field strength tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ transforms covariantly under Lorentz transformations:

$$F'_{\mu\nu} = \Lambda_\mu{}^\rho \Lambda_\nu{}^\sigma F_{\rho\sigma}.$$

Therefore, the scalar contraction $F_{\mu\nu}F^{\mu\nu}$ is invariant. Since the measure d^4x is also Lorentz invariant, the whole action $S[A]$ remains unchanged under Lorentz transformations.

Hence, $S[A]$ describes a **local, Lorentz-invariant field theory** for the electromagnetic field.

Quantum Field Theory – Homework 2 (Solutions)

Name: _____ Due Date: 2025.11.03

Problem 1. Probability Interpretation and the Klein–Gordon Equation (20 points)

In non-relativistic quantum mechanics, the Schrödinger equation

$$i \frac{\partial}{\partial t} \psi(\vec{x}, t) = -\frac{1}{2m} \nabla^2 \psi(\vec{x}, t)$$

admits a conserved quantity interpreted as probability. Multiplying by ψ^* and subtracting the complex conjugate gives

$$\frac{\partial}{\partial t} |\psi|^2 + \nabla \cdot \left[\frac{i}{2m} (\psi \nabla \psi^* - \psi^* \nabla \psi) \right] = 0.$$

Hence we can identify

$$\rho = |\psi|^2, \quad \vec{J} = \frac{i}{2m} (\psi \nabla \psi^* - \psi^* \nabla \psi),$$

which satisfy the continuity equation $\partial_t \rho + \nabla \cdot \vec{J} = 0$, so that ρ is real, non-negative, and represents the probability density.

Now consider the relativistic Klein–Gordon equation for a single relativistic particle

$$\partial_\mu \partial^\mu \psi(\vec{x}, t) - m^2 \psi(\vec{x}, t) = 0.$$

- (a) Derive the conserved current associated with this equation by combining it with its complex conjugate. Show that it can be written as

$$\partial_\mu j^\mu = 0, \quad j^\mu = i(\psi^* \partial^\mu \psi - \psi \partial^\mu \psi^*).$$

Identify the time component j^0 and spatial component $\vec{j}$.

Starting from the Klein–Gordon equation

$$(\partial_\mu \partial^\mu - m^2) \psi = 0,$$

and its complex conjugate

$$(\partial_\mu \partial^\mu - m^2) \psi^* = 0,$$

multiply the first by ψ^* and the second by ψ and subtract:

$$\psi^*(\partial_\mu\partial^\mu\psi) - \psi(\partial_\mu\partial^\mu\psi^*) = 0.$$

Rewriting gives

$$\partial_\mu(\psi^*\partial^\mu\psi - \psi\partial^\mu\psi^*) = 0.$$

Hence the conserved current is

$$\boxed{j^\mu = i(\psi^*\partial^\mu\psi - \psi\partial^\mu\psi^*) \quad \text{with} \quad \partial_\mu j^\mu = 0.}$$

The time component and spatial part are

$$j^0 = i(\psi^*\dot{\psi} - \psi\dot{\psi}^*), \quad \vec{j} = i(\psi^*\nabla\psi - \psi\nabla\psi^*).$$

- (b) Examine the expression for j^0 and discuss its sign. Why can j^0 take negative values even for well-behaved wave packets? Explain why it therefore cannot be interpreted as a probability density.

The quantity $j^0 = i(\psi^*\dot{\psi} - \psi\dot{\psi}^*)$ is not positive-definite. For plane-wave solutions $\psi \sim e^{-iEt+i\vec{k}\cdot\vec{x}}$, we have

$$j^0 = 2E|\psi|^2.$$

For positive-frequency modes $E > 0$, $j^0 > 0$, but for negative-frequency solutions $E < 0$, $j^0 < 0$. Hence, j^0 can take either sign and cannot be interpreted as a probability density.

- (c) Briefly explain the physical implication of this result. Why does the failure of a positive-definite probability density indicate that the Klein-Gordon equation cannot consistently describe a single relativistic particle, and motivates the need for a field-theoretic description?

Because j^0 is not positive definite, the Klein-Gordon equation cannot describe the probability density of a single relativistic particle. Instead, ψ must be interpreted as a **field**, not a wavefunction, whose excitations correspond to particles and antiparticles. This failure motivates the transition from relativistic quantum mechanics to **quantum field theory**.

Problem 2. Noethers Theorem and Energy-Momentum Conservation (20 points)

Consider the real scalar field theory

$$S[\phi] = \int d^4x \left[-\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi) \right].$$

- (a) Show that the action is invariant under spacetime translations $x^\mu \rightarrow x^\mu + a^\mu$.

We consider the real scalar field with Lagrangian density

$$\mathcal{L}(\phi, \partial_\mu \phi) = -\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi).$$

A spacetime translation is defined by

$$x^\mu \rightarrow x'^\mu = x^\mu + a^\mu, \quad a^\mu = \text{constant}.$$

A scalar field satisfies

$$\phi'(x') = \phi(x).$$

Expanding to first order in the infinitesimal parameter a^μ , we obtain

$$\phi'(x) = \phi(x - a) = \phi(x) - a^\nu \partial_\nu \phi(x),$$

so the field variation at fixed x is

$$\delta\phi(x) = \phi'(x) - \phi(x) = -a^\nu \partial_\nu \phi(x).$$

The measure is invariant under translations:

$$d^4x' = d^4x.$$

Since the Lagrangian contains no explicit dependence on x^μ ,

$$\mathcal{L}'(\phi', \partial\phi') = \mathcal{L}(\phi, \partial\phi).$$

Thus the action transforms as

$$S' = \int d^4x \mathcal{L}' = \int d^4x \mathcal{L} = S,$$

which shows that the action is invariant under spacetime translations.

- (b) Using Noethers theorem, derive the conserved current associated with this symmetry and identify the corresponding conserved quantity.

Under a general spacetime symmetry, both the field and coordinates vary:

$$\delta x^\mu = a^\mu, \quad \delta \phi = -a^\nu \partial_\nu \phi.$$

Derivation of the Noether current for spacetime symmetries. For space-time transformations, the variation of the action contains two pieces:

$$\delta S = \int d^4x (\delta \mathcal{L} + \mathcal{L} \partial_\mu \delta x^\mu).$$

The variation of the Lagrangian at fixed x is

$$\hat{\delta} \mathcal{L} = \frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \partial_\mu (\delta \phi).$$

Integrating the second term by parts and using the Euler–Lagrange equations gives

$$\delta S = \int d^4x \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta \phi + \mathcal{L} \delta x^\mu \right].$$

If the transformation is a symmetry, then $\delta S = 0$ for arbitrary integration regions, so the bracket must be a conserved current:

$$J^\mu = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta \phi + \mathcal{L} \delta x^\mu.$$

Applying this to translations. For a translation in the ν -direction:

$$a^\mu = \epsilon \delta^\mu{}_\nu, \quad \delta x^\mu = \epsilon \delta^\mu{}_\nu, \quad \delta \phi = -\epsilon \partial_\nu \phi.$$

Substituting:

$$J^\mu_{(\nu)} = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} (-\partial_\nu \phi) + \mathcal{L} \delta^\mu{}_\nu.$$

For our scalar Lagrangian,

$$\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} = -\partial^\mu \phi,$$

so

$$J^\mu_{(\nu)} = \partial^\mu \phi \partial_\nu \phi - \delta^\mu{}_\nu \mathcal{L}.$$

Thus the Noether current is

$$T^\mu{}_\nu = \partial^\mu \phi \partial_\nu \phi - \delta^\mu{}_\nu \mathcal{L}, \quad \partial_\mu T^\mu{}_\nu = 0.$$

The conserved charges are

$$Q_{(\nu)} = \int d^3x T^0{}_\nu.$$

These correspond to:

$$Q_{(0)} = \text{total energy}, \quad Q_{(i)} = \text{components of total momentum}.$$

- (c) Write down the components of the energy-momentum tensor $T^{\mu\nu}$ for this theory.

From the expression above,

$$T^\mu{}_\nu = \partial^\mu \phi \partial_\nu \phi - \delta^\mu{}_\nu \mathcal{L}.$$

Raising the second index:

$$T^{\mu\nu} = \partial^\mu \phi \partial^\nu \phi - g^{\mu\nu} \left[\frac{1}{2} (\partial_\alpha \phi)(\partial^\alpha \phi) + V(\phi) \right].$$

Explicitly:

$$T^{00} = \frac{1}{2} (\partial_0 \phi)^2 + \frac{1}{2} (\nabla \phi)^2 + V(\phi),$$

$$T^{0i} = \partial^0 \phi \partial^i \phi,$$

$$T^{ij} = \partial^i \phi \partial^j \phi - \delta^{ij} \left[\frac{1}{2} (\partial_\alpha \phi)(\partial^\alpha \phi) + V(\phi) \right].$$

Problem 3. Lorentz Invariance of the Scalar Field Action (15 points)

Show explicitly that the scalar field action

$$S = \int d^4x \left[-\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 \right]$$

is invariant under an infinitesimal Lorentz transformation

$$x^\mu \rightarrow x'^\mu = x^\mu + \Lambda^\mu{}_\nu x^\nu, \quad \phi(x) \rightarrow \phi'(x') = \phi(x),$$

where $\Lambda_{\mu\nu} = -\Lambda_{\nu\mu}$.

We want to show that

$$S[\phi] = \int d^4x \left[-\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 \right]$$

is invariant under the infinitesimal Lorentz transformation

$$x^\mu \rightarrow x'^\mu = x^\mu + \Lambda^\mu{}_\nu x^\nu, \quad \phi(x) \rightarrow \phi'(x') = \phi(x),$$

with $\Lambda_{\mu\nu} = -\Lambda_{\nu\mu}$ and $|\Lambda^\mu{}_\nu| \ll 1$.

1. Transformation of the measure. We first compute the Jacobian of the coordinate change:

$$\frac{\partial x'^\mu}{\partial x^\nu} = \delta^\mu{}_\nu + \Lambda^\mu{}_\nu.$$

Then

$$\det\left(\frac{\partial x'}{\partial x}\right) = \det(\delta^\mu{}_\nu + \Lambda^\mu{}_\nu) = 1 + \Lambda^\mu{}_\mu + \mathcal{O}(\Lambda^2).$$

Because $\Lambda_{\mu\nu}$ is antisymmetric,

$$\Lambda^\mu{}_\mu = \eta^{\mu\rho} \Lambda_{\rho\mu} = -\eta^{\mu\rho} \Lambda_{\mu\rho} = 0.$$

Hence

$$\det\left(\frac{\partial x'}{\partial x}\right) = 1 + \mathcal{O}(\Lambda^2),$$

so, to first order in Λ ,

$$d^4x' = d^4x.$$

Thus the measure is invariant at this order.

2. Transformation of the field and its derivatives. The field is a scalar, so

$$\phi'(x') = \phi(x).$$

For later convenience, it is useful to have the inverse transformation:

$$x^\mu = x'^\mu - \Lambda^\mu{}_\nu x'^\nu + \mathcal{O}(\Lambda^2).$$

Then

$$\phi'(x') = \phi(x) = \phi(x' - \Lambda x') = \phi(x') - \Lambda^\rho{}_\sigma x'^\sigma \partial_\rho \phi(x') + \mathcal{O}(\Lambda^2).$$

Therefore, at fixed x' ,

$$\delta\phi(x') \equiv \phi'(x') - \phi(x') = -\Lambda^\rho{}_\sigma x'^\sigma \partial_\rho \phi(x').$$

Next, the derivatives transform via the chain rule. The derivative with respect to x' is

$$\partial'_\mu \equiv \frac{\partial}{\partial x'^\mu} = \frac{\partial x^\rho}{\partial x'^\mu} \frac{\partial}{\partial x^\rho}.$$

Using

$$x^\rho = x'^\rho - \Lambda^\rho{}_\sigma x'^\sigma,$$

we obtain

$$\frac{\partial x^\rho}{\partial x'^\mu} = \delta^\rho{}_\mu - \Lambda^\rho{}_\mu,$$

so to first order

$$\partial'_\mu = (\delta^\rho{}_\mu - \Lambda^\rho{}_\mu) \partial_\rho.$$

Now act on the scalar field:

$$\partial'_\mu \phi'(x') = (\delta^\rho{}_\mu - \Lambda^\rho{}_\mu) \partial_\rho \phi(x).$$

To first order we can identify $x \approx x'$, so

$$\partial'_\mu \phi'(x') = (\delta^\rho{}_\mu - \Lambda^\rho{}_\mu) \partial_\rho \phi(x') + \mathcal{O}(\Lambda^2).$$

3. Invariance of the kinetic term. The kinetic term in the transformed action is

$$\partial'_\mu \phi'(x') \partial'^\mu \phi'(x') = \eta^{\mu\nu} \partial'_\mu \phi'(x') \partial'_\nu \phi'(x').$$

Substitute the expression for $\partial'_\mu \phi'$:

$$\begin{aligned}\partial'_\mu \phi' \partial'_\nu \phi' &= (\delta^\rho_\mu - \Lambda^\rho_\mu)(\delta^\sigma_\nu - \Lambda^\sigma_\nu) \partial_\rho \phi \partial_\sigma \phi \\ &= [\delta^\rho_\mu \delta^\sigma_\nu - \Lambda^\rho_\mu \delta^\sigma_\nu - \delta^\rho_\mu \Lambda^\sigma_\nu + \mathcal{O}(\Lambda^2)] \partial_\rho \phi \partial_\sigma \phi.\end{aligned}$$

Thus

$$\begin{aligned}\partial'_\mu \phi' \partial'^\mu \phi' &= \eta^{\mu\nu} \partial'_\mu \phi' \partial'_\nu \phi' \\ &= \eta^{\mu\nu} [\delta^\rho_\mu \delta^\sigma_\nu - \Lambda^\rho_\mu \delta^\sigma_\nu - \delta^\rho_\mu \Lambda^\sigma_\nu] \partial_\rho \phi \partial_\sigma \phi + \mathcal{O}(\Lambda^2) \\ &= \eta^{\rho\sigma} \partial_\rho \phi \partial_\sigma \phi - \eta^{\mu\nu} \Lambda^\rho_\mu \delta^\sigma_\nu \partial_\rho \phi \partial_\sigma \phi - \eta^{\mu\nu} \delta^\rho_\mu \Lambda^\sigma_\nu \partial_\rho \phi \partial_\sigma \phi + \mathcal{O}(\Lambda^2).\end{aligned}$$

We now simplify the second and third terms. Using

$$\eta^{\mu\nu} \Lambda^\rho_\mu \delta^\sigma_\nu = \Lambda^{\rho\sigma}, \quad \eta^{\mu\nu} \delta^\rho_\mu \Lambda^\sigma_\nu = \Lambda^{\sigma\rho},$$

we get

$$\partial'_\mu \phi' \partial'^\mu \phi' = \partial_\rho \phi \partial^\rho \phi - \Lambda^{\rho\sigma} \partial_\rho \phi \partial_\sigma \phi - \Lambda^{\sigma\rho} \partial_\rho \phi \partial_\sigma \phi + \mathcal{O}(\Lambda^2).$$

But $\Lambda^{\rho\sigma} = -\Lambda^{\sigma\rho}$ is antisymmetric, while $\partial_\rho \phi \partial_\sigma \phi$ is symmetric in (ρ, σ) . Therefore the contractions vanish:

$$\Lambda^{\rho\sigma} \partial_\rho \phi \partial_\sigma \phi = 0, \quad \Lambda^{\sigma\rho} \partial_\rho \phi \partial_\sigma \phi = 0.$$

So we obtain

$$\partial'_\mu \phi' \partial'^\mu \phi' = \partial_\mu \phi \partial^\mu \phi + \mathcal{O}(\Lambda^2).$$

Thus the kinetic term is invariant to first order in Λ .

4. Invariance of the mass term. Since $\phi'(x') = \phi(x)$, we have

$$\phi'^2(x') = \phi^2(x),$$

so the mass term $m^2 \phi^2$ is clearly invariant:

$$m^2 \phi'^2(x') = m^2 \phi^2(x).$$

5. Putting it all together. The transformed action is

$$S' = \int d^4 x' \left[-\frac{1}{2} \partial'_\mu \phi' \partial'^\mu \phi' - \frac{1}{2} m^2 \phi'^2 \right].$$

Changing variables back to x and using $d^4x' = d^4x$ (to first order), and the invariance of both the kinetic and mass terms, we get

$$S' = \int d^4x \left[-\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 \right] = S.$$

Therefore $\delta S = S' - S = 0$ to first order in Λ , and the scalar field action is invariant under infinitesimal Lorentz transformations.

Then, use Noether's theorem to find the expression for the conserved angular-momentum current $M^{\lambda\mu\nu}$.

We now derive the conserved current associated with Lorentz invariance and show that it can be written as

$$M^{\lambda\mu\nu} = x^\mu T^{\lambda\nu} - x^\nu T^{\lambda\mu}.$$

1. Infinitesimal Lorentz transformation. An infinitesimal Lorentz transformation is

$$x'^\mu = x^\mu + \omega^\mu{}_\nu x^\nu, \quad \omega_{\mu\nu} = -\omega_{\nu\mu},$$

where $\omega_{\mu\nu}$ are small, antisymmetric parameters. For a scalar field, we have

$$\phi'(x') = \phi(x).$$

As before, expanding to first order gives the field variation at fixed x :

$$\delta\phi(x) \equiv \phi'(x) - \phi(x) = -\omega^\rho{}_\sigma x^\sigma \partial_\rho \phi(x).$$

Similarly,

$$\delta x^\mu = \omega^\mu{}_\nu x^\nu.$$

2. Noether current for spacetime symmetries. From our general result for spacetime symmetries, the Noether current is

$$J^\lambda = \frac{\partial \mathcal{L}}{\partial(\partial_\lambda \phi)} \delta\phi + \mathcal{L} \delta x^\lambda.$$

For the scalar field Lagrangian

$$\mathcal{L} = -\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2,$$

the momentum conjugate to $\partial_\lambda \phi$ is

$$\frac{\partial \mathcal{L}}{\partial(\partial_\lambda \phi)} = -\partial^\lambda \phi.$$

Insert the Lorentz variations:

$$\begin{aligned} J^\lambda &= (-\partial^\lambda \phi) (-\omega^\rho{}_\sigma x^\sigma \partial_\rho \phi) + \mathcal{L} \omega^\lambda{}_\sigma x^\sigma \\ &= \omega^\rho{}_\sigma x^\sigma \partial^\lambda \phi \partial_\rho \phi + \omega^\lambda{}_\sigma x^\sigma \mathcal{L}. \end{aligned}$$

3. Isolating the current for each generator. Because $\omega_{\mu\nu}$ is antisymmetric, it is convenient to rewrite $\omega^\rho{}_\sigma$ in terms of $\omega_{\mu\nu}$ and then factor out the independent parameters. We use

$$\omega^\rho{}_\sigma = \eta^{\rho\alpha} \omega_{\alpha\sigma},$$

and write the current as a linear combination of $\omega_{\mu\nu}$:

$$J^\lambda = \frac{1}{2} \omega_{\mu\nu} M^{\lambda\mu\nu},$$

for some tensor $M^{\lambda\mu\nu}$ antisymmetric in (μ, ν) . To find $M^{\lambda\mu\nu}$, note that a general antisymmetric $\omega_{\mu\nu}$ can be written as a sum over basis matrices that generate boosts and rotations. Comparing

$$J^\lambda = \omega^\rho{}_\sigma (x^\sigma \partial^\lambda \phi \partial_\rho \phi) + \omega^\lambda{}_\sigma x^\sigma \mathcal{L}$$

with the general form $\frac{1}{2} \omega_{\mu\nu} M^{\lambda\mu\nu}$, we can guess the structure: it must be linear in x times the energy-momentum tensor $T^\lambda{}_\rho$. Recall the canonical energy-momentum tensor obtained previously,

$$T^\lambda{}_\rho = \partial^\lambda \phi \partial_\rho \phi - \delta^\lambda{}_\rho \mathcal{L}.$$

Consider the combination

$$M^{\lambda\mu\nu} = x^\mu T^{\lambda\nu} - x^\nu T^{\lambda\mu}.$$

This tensor is antisymmetric in (μ, ν) by construction:

$$M^{\lambda\nu\mu} = -M^{\lambda\mu\nu}.$$

Now form the current with $\omega_{\mu\nu}$:

$$\begin{aligned} J^\lambda &= \frac{1}{2} \omega_{\mu\nu} M^{\lambda\mu\nu} \\ &= \frac{1}{2} \omega_{\mu\nu} (x^\mu T^{\lambda\nu} - x^\nu T^{\lambda\mu}). \end{aligned}$$

Using the antisymmetry $\omega_{\mu\nu} = -\omega_{\nu\mu}$ and relabeling dummy indices, one can check that this reproduces the previous expression for J^λ . Thus we identify

$$\boxed{M^{\lambda\mu\nu} = x^\mu T^{\lambda\nu} - x^\nu T^{\lambda\mu}.}$$

4. Interpretation and conservation. Noethers theorem gives the conservation law

$$\partial_\lambda M^{\lambda\mu\nu} = 0,$$

which expresses conservation of total (orbital + spin) angular momentum associated with Lorentz invariance. In general, for fields with spin, there can be an extra “spin current” term $S^{\lambda\mu\nu}$ so that

$$M^{\lambda\mu\nu} = x^\mu T^{\lambda\nu} - x^\nu T^{\lambda\mu} + S^{\lambda\mu\nu}.$$

For a *real scalar field*, there is no intrinsic spin, so $S^{\lambda\mu\nu} = 0$, and the angular-momentum current is purely *orbital*, given exactly by

$$M^{\lambda\mu\nu} = x^\mu T^{\lambda\nu} - x^\nu T^{\lambda\mu}.$$

The corresponding conserved charges are

$$J^{\mu\nu} = \int d^3x M^{0\mu\nu}(t, \vec{x}),$$

which generate Lorentz transformations (rotations and boosts) on the fields. In particular, the spatial components J^{ij} are the usual angular momentum operators, while J^{0i} generate boosts.

Problem 4. From Discrete Systems to Field Theory (15 points)

Starting from the Lagrangian of a one-dimensional chain of coupled oscillators

$$L = \sum_n \left[\frac{1}{2} \mu \dot{\eta}_n^2 - \frac{\lambda}{2} (\eta_{n+1} - \eta_n)^2 - \frac{1}{2} \sigma \eta_n^2 \right],$$

derive its continuum limit as the lattice spacing $a \rightarrow 0$, obtaining a field theory in terms of a continuous field $\eta(x, t)$. Identify the effective mass m and propagation speed v , and discuss under what condition this theory becomes relativistic.

Start from

$$L = \sum_n \left[\frac{1}{2} \mu \dot{\eta}_n^2 - \frac{\lambda}{2} (\eta_{n+1} - \eta_n)^2 - \frac{1}{2} \sigma \eta_n^2 \right].$$

Let $x = na$ and $\eta_n(t) \rightarrow \eta(x, t)$. For small lattice spacing a ,

$$\eta_{n+1} - \eta_n \approx a \partial_x \eta,$$

and the sum becomes an integral

$$\sum_n \rightarrow \frac{1}{a} \int dx.$$

Then

$$L = \int dx \left[\frac{1}{2} \mu (\partial_t \eta)^2 - \frac{1}{2} \lambda a^2 (\partial_x \eta)^2 - \frac{1}{2} \sigma \eta^2 \right].$$

Define the continuum field theory Lagrangian density

$$\mathcal{L} = \frac{1}{2} \mu (\partial_t \eta)^2 - \frac{1}{2} \lambda a^2 (\partial_x \eta)^2 - \frac{1}{2} \sigma \eta^2.$$

We can identify

$$v^2 = \frac{\lambda a^2}{\mu}, \quad m^2 = \frac{\sigma}{\mu}.$$

Hence, in the continuum limit,

$$\boxed{\mathcal{L} = \frac{1}{2} (\partial_t \eta)^2 - \frac{1}{2} v^2 (\partial_x \eta)^2 - \frac{1}{2} m^2 \eta^2.}$$

The equation of motion is

$$\partial_t^2 \eta - v^2 \partial_x^2 \eta + m^2 \eta = 0.$$

The theory becomes **relativistic** when $v = 1$, so that time and space derivatives enter symmetrically.

Quantum Field Theory – Homework 3

Name: _____ Due Date: 2025.11.10

The following problems are based on Lectures 5 and 6, focusing on the quantization of the harmonic oscillator and the free scalar field. The bullet points are guiding hints to help structure your answers.

Problem 1. Quantization of the Harmonic Oscillator (20 points)

Consider the one-dimensional harmonic oscillator with Lagrangian

$$L = \frac{1}{2}\dot{x}^2 - \frac{1}{2}\omega^2 x^2.$$

- (a) Derive the conjugate momentum and Hamiltonian of the system.

The Lagrangian is

$$L = \frac{1}{2}\dot{x}^2 - \frac{1}{2}\omega^2 x^2.$$

The momentum conjugate to x is

$$p = \frac{\partial L}{\partial \dot{x}} = \dot{x}.$$

The Hamiltonian is the Legendre transform:

$$H = p\dot{x} - L.$$

Since $\dot{x} = p$, we obtain

$$H = p^2 - \left(\frac{1}{2}p^2 - \frac{1}{2}\omega^2 x^2\right) = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 x^2.$$

- (b) Quantize the system by promoting x and p to operators satisfying the canonical commutation relation $[\hat{x}, \hat{p}] = i$.

Promote x, p to operators obeying the canonical commutation relation

$$[\hat{x}, \hat{p}] = i.$$

The Hamiltonian becomes

$$\hat{H} = \frac{1}{2}\hat{p}^2 + \frac{1}{2}\omega^2 \hat{x}^2.$$

(c) Define the creation and annihilation operators

$$\hat{a} = \frac{1}{\sqrt{2\omega}}(\omega\hat{x} + i\hat{p}), \quad \hat{a}^\dagger = \frac{1}{\sqrt{2\omega}}(\omega\hat{x} - i\hat{p}),$$

and derive their commutation relation.

Define

$$\hat{a} = \frac{1}{\sqrt{2\omega}}(\omega\hat{x} + i\hat{p}), \quad \hat{a}^\dagger = \frac{1}{\sqrt{2\omega}}(\omega\hat{x} - i\hat{p}).$$

Compute the commutator:

$$[\hat{a}, \hat{a}^\dagger] = \frac{1}{2\omega}[\omega\hat{x} + i\hat{p}, \omega\hat{x} - i\hat{p}].$$

Using bilinearity and

$$[\hat{x}, \hat{p}] = i, \quad [\hat{p}, \hat{x}] = -i,$$

we obtain

$$[\hat{a}, \hat{a}^\dagger] = \frac{1}{2\omega}(-i\omega[\hat{x}, \hat{p}] + i\omega[\hat{p}, \hat{x}]) = \frac{1}{2\omega}(-i\omega i + i\omega(-i)) = 1.$$

Thus

$$[\hat{a}, \hat{a}^\dagger] = 1.$$

(d) Express the Hamiltonian in terms of $\hat{a}$ and $\hat{a}^\dagger$, and find the general form of its energy eigenvalues and eigenstates.

Solve for $\hat{x}$ and $\hat{p}$:

$$\hat{x} = \frac{1}{\sqrt{2\omega}}(\hat{a} + \hat{a}^\dagger), \quad \hat{p} = \frac{\sqrt{\omega}}{\sqrt{2}}\frac{1}{i}(\hat{a} - \hat{a}^\dagger) = -i\sqrt{\frac{\omega}{2}}(\hat{a} - \hat{a}^\dagger).$$

Insert into

$$\hat{H} = \frac{1}{2}\hat{p}^2 + \frac{1}{2}\omega^2\hat{x}^2.$$

After simplification, using $[\hat{a}, \hat{a}^\dagger] = 1$, one finds

$$\hat{H} = \omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right).$$

Therefore the energy eigenvalues are

$$E_n = \omega \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, \dots$$

with eigenstates

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}} |0\rangle,$$

where the ground state satisfies $\hat{a}|0\rangle = 0$.

Problem 2. Quantization of the Free Scalar Field (25 points)

Consider the free real scalar field with action

$$S = \int d^4x \left[\frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2}m^2 \phi^2 \right].$$

- (a) Derive the field equation of motion and the conjugate momentum.

The action is

$$S[\phi] = \int d^4x \left[\frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2}m^2 \phi^2 \right].$$

The Lagrangian density is

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2}m^2 \phi^2.$$

The Euler-Lagrange equation for a field is

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0.$$

We compute

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = \partial^\mu \phi, \quad \frac{\partial \mathcal{L}}{\partial \phi} = -m^2 \phi.$$

Thus

$$\partial_\mu \partial^\mu \phi + m^2 \phi = 0,$$

which is the Klein-Gordon equation. The conjugate momentum is

$$\Pi(x) \equiv \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi)} = \partial^0 \phi = \dot{\phi}(t, \vec{x}).$$

- (b) Write down the general mode expansion of the quantum field operator $\hat{\phi}(x)$ in terms of creation and annihilation operators $\hat{a}_{\vec{k}}$ and $\hat{a}_{\vec{k}}^\dagger$.

We promote ϕ and Π to operators $\hat{\phi}$ and $\hat{\Pi}$. A general solution of the Klein-Gordon equation can be written as a superposition of plane waves with on-shell frequency

$$\omega_{\vec{k}} = \sqrt{\vec{k}^2 + m^2}.$$

The standard mode expansion of the real scalar field operator is

$$\hat{\phi}(x) = \hat{\phi}(t, \vec{x}) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \left[\hat{a}_{\vec{k}} e^{-i\omega_{\vec{k}}t + i\vec{k} \cdot \vec{x}} + \hat{a}_{\vec{k}}^\dagger e^{+i\omega_{\vec{k}}t - i\vec{k} \cdot \vec{x}} \right]. \quad (1)$$

For later use we also note the expansion of the conjugate momentum:

$$\hat{\Pi}(t, \vec{x}) = \dot{\hat{\phi}}(t, \vec{x}) = -i \int \frac{d^3 k}{(2\pi)^3} \sqrt{\frac{\omega_{\vec{k}}}{2}} \left[\hat{a}_{\vec{k}} e^{-i\omega_{\vec{k}} t + i\vec{k} \cdot \vec{x}} - \hat{a}_{\vec{k}}^\dagger e^{+i\omega_{\vec{k}} t - i\vec{k} \cdot \vec{x}} \right]. \quad (2)$$

- (c) Starting from the canonical commutation relation $[\hat{\phi}(t, \vec{x}), \hat{\Pi}(t, \vec{x}')] = i\delta^{(3)}(\vec{x} - \vec{x}')$, derive the commutator between $\hat{a}_{\vec{k}}$ and $\hat{a}_{\vec{k}'}^\dagger$.

We impose the equal-time canonical commutation relations

$$[\hat{\phi}(t, \vec{x}), \hat{\Pi}(t, \vec{x}')] = i\delta^{(3)}(\vec{x} - \vec{x}'), \quad [\hat{\phi}(t, \vec{x}), \hat{\phi}(t, \vec{x}')] = 0, \quad [\hat{\Pi}(t, \vec{x}), \hat{\Pi}(t, \vec{x}')] = 0.$$

To extract the commutator of the modes, it is convenient to invert the mode expansion. Using the completeness of plane waves, one finds

$$\hat{a}_{\vec{k}} = \int d^3 x \left[\sqrt{\frac{\omega_{\vec{k}}}{2}} \hat{\phi}(t, \vec{x}) + \frac{i}{\sqrt{2\omega_{\vec{k}}}} \hat{\Pi}(t, \vec{x}) \right] e^{-i\vec{k} \cdot \vec{x}}, \quad (3)$$

$$\hat{a}_{\vec{k}}^\dagger = \int d^3 x \left[\sqrt{\frac{\omega_{\vec{k}}}{2}} \hat{\phi}(t, \vec{x}) - \frac{i}{\sqrt{2\omega_{\vec{k}}}} \hat{\Pi}(t, \vec{x}) \right] e^{+i\vec{k} \cdot \vec{x}}. \quad (4)$$

Now compute

$$[\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}'}^\dagger] = \int d^3 x d^3 x' e^{-i\vec{k} \cdot \vec{x}} e^{+i\vec{k}' \cdot \vec{x}'} \left[\sqrt{\frac{\omega_{\vec{k}}}{2}} \hat{\phi}(t, \vec{x}) + \frac{i}{\sqrt{2\omega_{\vec{k}}}} \hat{\Pi}(t, \vec{x}), \sqrt{\frac{\omega_{\vec{k}'}}{2}} \hat{\phi}(t, \vec{x}') - \frac{i}{\sqrt{2\omega_{\vec{k}'}}} \hat{\Pi}(t, \vec{x}') \right].$$

Only the cross terms containing $[\hat{\phi}(t, \vec{x}), \hat{\Pi}(t, \vec{x}')] survive:$

$$\begin{aligned} [\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}'}^\dagger] &= \int d^3 x d^3 x' e^{-i\vec{k} \cdot \vec{x}} e^{+i\vec{k}' \cdot \vec{x}'} \left[\sqrt{\frac{\omega_{\vec{k}}}{2}} \left(-\frac{i}{\sqrt{2\omega_{\vec{k}'}}} \right) [\hat{\phi}(t, \vec{x}), \hat{\Pi}(t, \vec{x}')] \right. \\ &\quad \left. + \frac{i}{\sqrt{2\omega_{\vec{k}}}} \sqrt{\frac{\omega_{\vec{k}'}}{2}} [\hat{\Pi}(t, \vec{x}), \hat{\phi}(t, \vec{x}')] \right]. \end{aligned}$$

Using

$$[\hat{\phi}(t, \vec{x}), \hat{\Pi}(t, \vec{x}')] = i\delta^{(3)}(\vec{x} - \vec{x}'), \quad [\hat{\Pi}(t, \vec{x}), \hat{\phi}(t, \vec{x}')] = -i\delta^{(3)}(\vec{x} - \vec{x}'),$$

we get

$$\begin{aligned}
 [\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}'}^\dagger] &= \int d^3x d^3x' e^{-i\vec{k}\cdot\vec{x}} e^{+i\vec{k}'\cdot\vec{x}'} \left[\sqrt{\frac{\omega_{\vec{k}}}{2}} \left(-\frac{i}{\sqrt{2\omega_{\vec{k}'}}} \right) i\delta^{(3)}(\vec{x} - \vec{x}') + \frac{i}{\sqrt{2\omega_{\vec{k}}}} \sqrt{\frac{\omega_{\vec{k}'}}{2}} (-i)\delta^{(3)}(\vec{x} - \vec{x}') \right] \\
 &= \int d^3x e^{-i\vec{k}\cdot\vec{x}} e^{+i\vec{k}'\cdot\vec{x}} \left[\frac{\omega_{\vec{k}}}{2\sqrt{\omega_{\vec{k}}}\omega_{\vec{k}'}} + \frac{\omega_{\vec{k}'}}{2\sqrt{\omega_{\vec{k}}}\omega_{\vec{k}'}} \right] \\
 &= \int d^3x e^{i(\vec{k}' - \vec{k})\cdot\vec{x}} = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}').
 \end{aligned}$$

We also find

$$[\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}'}] = 0, \quad [\hat{a}_{\vec{k}}^\dagger, \hat{a}_{\vec{k}'}^\dagger] = 0.$$

Hence

$$[\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}'}^\dagger] = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}'), \quad [\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}'}] = 0 = [\hat{a}_{\vec{k}}^\dagger, \hat{a}_{\vec{k}'}^\dagger].$$

- (d) Explain why each mode labeled by $\vec{k}$ behaves as an independent harmonic oscillator, and describe the physical meaning of $\hat{a}_{\vec{k}}^\dagger$ acting on the vacuum state.

The Hamiltonian density for the free scalar field is

$$\mathcal{H} = \frac{1}{2}\hat{\Pi}^2 + \frac{1}{2}(\nabla\hat{\phi})^2 + \frac{1}{2}m^2\hat{\phi}^2,$$

and the Hamiltonian is

$$\hat{H} = \int d^3x \mathcal{H}.$$

Inserting the mode expansions (1) and (2) and using the commutation relations, one finds (after normal ordering and discarding the infinite c-number vacuum energy)

$$\hat{H} = \int \frac{d^3k}{(2\pi)^3} \omega_{\vec{k}} \hat{a}_{\vec{k}}^\dagger \hat{a}_{\vec{k}} + (\text{constant}).$$

This has the same structure as a sum over independent harmonic oscillators: for each momentum $\vec{k}$ we have a mode with frequency $\omega_{\vec{k}}$ and number operator

$$\hat{N}_{\vec{k}} = \hat{a}_{\vec{k}}^\dagger \hat{a}_{\vec{k}},$$

contributing an energy $\omega_{\vec{k}}(N_{\vec{k}} + \frac{1}{2})$. Therefore each *Fourier mode* of the field behaves as an independent harmonic oscillator. The vacuum state $|\Omega\rangle$ is defined by

$$\hat{a}_{\vec{k}}|\Omega\rangle = 0 \quad \forall \vec{k}.$$

Acting with $\hat{a}_{\vec{k}}^\dagger$ on the vacuum creates a one-particle state with definite momentum $\vec{k}$:

$$|\vec{k}\rangle \equiv \hat{a}_{\vec{k}}^\dagger |\Omega\rangle.$$

This state is an eigenstate of the total momentum operator with eigenvalue $\vec{k}$ and of the Hamiltonian with energy $\omega_{\vec{k}}$. Thus, $\hat{a}_{\vec{k}}^\dagger |\Omega\rangle$ corresponds to a single scalar particle with momentum $\vec{k}$ and energy $\omega_{\vec{k}}$.

Problem 3. Zero-Point Energy and Momentum Cutoff (20 points)

The vacuum energy density of the free scalar field is formally given by

$$\epsilon_0 = \frac{1}{2} \int \frac{d^3 \vec{k}}{(2\pi)^3} \sqrt{\vec{k}^2 + m^2}.$$

- (a) Introduce a momentum cutoff Λ and evaluate the leading dependence of ϵ_0 on Λ in the limit $\Lambda \gg m$.

Introduce a sharp UV cutoff $|\vec{k}| \leq \Lambda$ and go to spherical coordinates in momentum space:

$$d^3 \vec{k} = 4\pi k^2 dk.$$

Then

$$\epsilon_0(\Lambda) = \frac{1}{2} \int_{|\vec{k}| \leq \Lambda} \frac{d^3 \vec{k}}{(2\pi)^3} \sqrt{\vec{k}^2 + m^2} \quad (5)$$

$$= \frac{1}{2} \frac{4\pi}{(2\pi)^3} \int_0^\Lambda k^2 \sqrt{k^2 + m^2} dk \quad (6)$$

$$= \frac{1}{4\pi^2} \int_0^\Lambda k^2 \sqrt{k^2 + m^2} dk. \quad (7)$$

For large k (i.e. in the UV region that dominates when $\Lambda \gg m$), expand the integrand:

$$\sqrt{k^2 + m^2} = k \sqrt{1 + \frac{m^2}{k^2}} \simeq k \left(1 + \frac{m^2}{2k^2} - \frac{m^4}{8k^4} + \dots \right).$$

Hence

$$k^2 \sqrt{k^2 + m^2} \simeq k^3 + \frac{1}{2} m^2 k - \frac{1}{8} \frac{m^4}{k} + \dots.$$

Integrating term by term up to Λ gives

$$\int_0^\Lambda k^3 dk = \frac{\Lambda^4}{4}, \quad (8)$$

$$\int_0^\Lambda k dk = \frac{\Lambda^2}{2}, \quad (9)$$

$$\int_0^\Lambda \frac{dk}{k} \sim \ln \frac{\Lambda}{m} \quad (\text{the lower limit is effectively of order } m). \quad (10)$$

Therefore

$$\epsilon_0(\Lambda) \simeq \frac{1}{4\pi^2} \left[\frac{\Lambda^4}{4} + \frac{m^2}{2} \frac{\Lambda^2}{2} - \frac{m^4}{8} \ln \frac{\Lambda}{m} + \dots \right], \quad (11)$$

or more neatly

$$\epsilon_0(\Lambda) \simeq \frac{\Lambda^4}{16\pi^2} + \frac{m^2\Lambda^2}{16\pi^2} - \frac{m^4}{32\pi^2} \ln \frac{\Lambda}{m} + \dots, \quad \Lambda \gg m. \quad (12)$$

The most important point is the quartic and quadratic divergences in Λ .

- (b) Explain the physical origin of the divergence in ϵ_0 .

In quantum field theory, each momentum mode $\vec{k}$ of the field is a harmonic oscillator with frequency

$$\omega_{\vec{k}} = \sqrt{\vec{k}^2 + m^2}.$$

Even in the vacuum, each oscillator contributes a zero-point energy $\frac{1}{2}\omega_{\vec{k}}$. The total vacuum energy density is the sum (over modes) of these $\frac{1}{2}\omega_{\vec{k}}$ contributions:

$$\epsilon_0 \sim \int \frac{d^3\vec{k}}{(2\pi)^3} \frac{1}{2}\omega_{\vec{k}}.$$

Because there are infinitely many modes and the number of modes grows like k^2 at large k , the integral over momenta diverges at high $|\vec{k}|$. This reflects the fact that a continuum field has an infinite number of degrees of freedom per unit volume, and each degree of freedom contributes a non-zero vacuum fluctuation energy.

- (c) Discuss briefly how this divergence relates to the continuum nature of field theory and why it leads to the so-called cosmological constant problem.

The divergence of ϵ_0 is directly tied to the continuum nature of field theory: momenta take continuous values up to arbitrarily large $|\vec{k}|$, so the density of modes is unbounded. Introducing a cutoff Λ (for example, a physical UV scale such as the Planck scale) yields a finite but typically enormous vacuum energy density:

$$\epsilon_0(\Lambda) \sim \frac{\Lambda^4}{16\pi^2} + \dots$$

In general relativity, vacuum energy gravitates and acts like a cosmological constant. Naively plugging the QFT estimate for ϵ_0 (with a high cutoff such as the Planck scale) into Einsteins equations gives a cosmological constant many orders of magnitude larger than the tiny value inferred from cosmological observations. Reconciling this huge theoretical vacuum energy with the observed small cosmological constant is the *cosmological constant problem*: it appears to require an extreme fine-tuning (or a new mechanism) to cancel the zero-point contributions almost exactly while leaving a small residual value that drives the accelerated expansion of the universe.

Quantum Field Theory – Homework 4

Name: _____ Due Date: 2025.11.17

The following problems are based on Lectures 7 and 8, focusing on the Hilbert space of the free scalar field and complex scalar field. The bullet points are guiding hints to help structure your answers.

Problem 1. Hamiltonian and Momentum Operators in Free Scalar Field Theory (20 points)

Consider a real free scalar field of mass m in $(3+1)$ dimensions with Lagrangian density

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2.$$

In the Heisenberg picture, the quantized field operator may be expanded as

$$\hat{\phi}(x) = \int \frac{d^3\vec{k}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \left(\hat{a}_{\vec{k}} e^{-i\omega_{\vec{k}}t + i\vec{k}\cdot\vec{x}} + \hat{a}_{\vec{k}}^\dagger e^{i\omega_{\vec{k}}t - i\vec{k}\cdot\vec{x}} \right),$$

where $\omega_{\vec{k}} = \sqrt{\vec{k}^2 + m^2}$.

(a) Derive the conjugate momentum field

$$\Pi(x) = \frac{\partial\mathcal{L}}{\partial(\partial_0\phi(x))}.$$

to have the conjugate momentum fields operator $\hat{\Pi}(x)$. Then, write down the equal-time canonical commutation relation between $\hat{\phi}(t, \vec{x})$ and $\hat{\Pi}(t, \vec{x}')$.

We start from the Lagrangian density

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2.$$

Using the metric convention $\eta_{\mu\nu} = \text{diag}(-, +, +, +)$, we can write

$$\partial_\mu\phi\partial^\mu\phi = \eta^{\mu\nu}\partial_\mu\phi\partial_\nu\phi = -(\partial_0\phi)^2 + (\nabla\phi)^2.$$

Therefore, the Lagrangian density becomes

$$\begin{aligned} \mathcal{L} &= -\frac{1}{2}[-(\partial_0\phi)^2 + (\nabla\phi)^2] - \frac{1}{2}m^2\phi^2 \\ &= \frac{1}{2}(\partial_0\phi)^2 - \frac{1}{2}(\nabla\phi)^2 - \frac{1}{2}m^2\phi^2. \end{aligned} \tag{1}$$

Conjugate momentum. By definition, the conjugate momentum field is

$$\Pi(x) = \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi(x))}.$$

From (1), the only term that depends on $\partial_0 \phi$ is $\frac{1}{2}(\partial_0 \phi)^2$, hence the classical conjugate momentum field is

$$\Pi(t, \vec{x}) = \frac{\partial}{\partial(\partial_0 \phi)} \left[\frac{1}{2}(\partial_0 \phi)^2 \right] = \partial_0 \phi(t, \vec{x}).$$

Therefore, the conjugate momentum operator is

$$\hat{\Pi}(t, \vec{x}) = \partial_0 \hat{\phi}(t, \vec{x}).$$

Mode expansion of the conjugate momentum. Starting from the mode expansion of the field operator,

$$\hat{\phi}(t, \vec{x}) = \int \frac{d^3 \vec{k}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \left(\hat{a}_{\vec{k}} e^{-i\omega_{\vec{k}} t + i\vec{k} \cdot \vec{x}} + \hat{a}_{\vec{k}}^\dagger e^{i\omega_{\vec{k}} t - i\vec{k} \cdot \vec{x}} \right),$$

we differentiate with respect to time to obtain the conjugate momentum operator:

$$\begin{aligned} \hat{\Pi}(t, \vec{x}) &= \partial_0 \hat{\phi}(t, \vec{x}) = \int \frac{d^3 \vec{k}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \left(-i\omega_{\vec{k}} \hat{a}_{\vec{k}} e^{-i\omega_{\vec{k}} t + i\vec{k} \cdot \vec{x}} + i\omega_{\vec{k}} \hat{a}_{\vec{k}}^\dagger e^{i\omega_{\vec{k}} t - i\vec{k} \cdot \vec{x}} \right) \\ &= \int \frac{d^3 \vec{k}}{(2\pi)^3} \sqrt{\frac{\omega_{\vec{k}}}{2}} \left(-i \hat{a}_{\vec{k}} e^{-i\omega_{\vec{k}} t + i\vec{k} \cdot \vec{x}} + i \hat{a}_{\vec{k}}^\dagger e^{i\omega_{\vec{k}} t - i\vec{k} \cdot \vec{x}} \right). \end{aligned} \quad (2)$$

Equal-time canonical commutation relations. Canonical quantization postulates the equal-time commutator

$$[\hat{\phi}(t, \vec{x}), \hat{\Pi}(t, \vec{x}')] = i \delta^{(3)}(\vec{x} - \vec{x}').$$

This relation is the quantum analogue of the classical Poisson bracket

$$\{\phi(t, \vec{x}), \Pi(t, \vec{x}')\} = \delta^{(3)}(\vec{x} - \vec{x}'),$$

together with the quantization rule

$$\{A, B\} \longrightarrow \frac{1}{i} [\hat{A}, \hat{B}].$$

In addition, the equal-time commutators among fields and among conjugate momenta vanish:

$$[\hat{\phi}(t, \vec{x}), \hat{\phi}(t, \vec{x}')] = 0, \quad [\hat{\Pi}(t, \vec{x}), \hat{\Pi}(t, \vec{x}')] = 0.$$

Then, substituting the mode expansions of $\hat{\phi}$ and $\hat{\Pi}$ into the canonical commutation relations fixes the algebra of the creation and annihilation operators:

$$[\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}'}^\dagger] = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}'), \quad [\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}'}] = [\hat{a}_{\vec{k}}^\dagger, \hat{a}_{\vec{k}'}^\dagger] = 0.$$

This shows that the mode expansion is consistent with canonical quantization.

(b) Starting from the Hamiltonian density

$$\hat{\mathcal{H}} = \frac{1}{2} \hat{\Pi}^2 + \frac{1}{2} (\nabla \hat{\phi})^2 + \frac{1}{2} m^2 \hat{\phi}^2,$$

construct the Hamiltonian operator

$$\hat{H} = \int d^3 \vec{x} \hat{\mathcal{H}}.$$

Rewrite $\hat{H}$ in terms of the creation and annihilation operators $\hat{a}_{\vec{k}}$ and $\hat{a}_{\vec{k}}^\dagger$.

The plane-wave expansion (at fixed time t) of the field operator $\hat{\phi}(t, \vec{x})$ is

$$\hat{\phi}(t, \vec{x}) = \int \frac{d^3 \vec{k}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \left(\hat{a}_{\vec{k}} e^{-i\omega_{\vec{k}}t + i\vec{k} \cdot \vec{x}} + \hat{a}_{\vec{k}}^\dagger e^{i\omega_{\vec{k}}t - i\vec{k} \cdot \vec{x}} \right),$$

where $\omega_{\vec{k}} = \sqrt{\vec{k}^2 + m^2}$. The conjugate momentum field operator is defined by

$$\hat{\Pi}(t, \vec{x}) = \partial_0 \hat{\phi}(t, \vec{x}),$$

which gives

$$\hat{\Pi}(t, \vec{x}) = \int \frac{d^3 \vec{k}}{(2\pi)^3} \sqrt{\frac{\omega_{\vec{k}}}{2}} \left(-i \hat{a}_{\vec{k}} e^{-i\omega_{\vec{k}}t + i\vec{k} \cdot \vec{x}} + i \hat{a}_{\vec{k}}^\dagger e^{i\omega_{\vec{k}}t - i\vec{k} \cdot \vec{x}} \right).$$

We will also need the spatial gradient of the field,

$$\nabla \hat{\phi}(t, \vec{x}) = \int \frac{d^3 \vec{k}}{(2\pi)^3} \frac{i\vec{k}}{\sqrt{2\omega_{\vec{k}}}} \left(\hat{a}_{\vec{k}} e^{-i\omega_{\vec{k}}t + i\vec{k} \cdot \vec{x}} - \hat{a}_{\vec{k}}^\dagger e^{i\omega_{\vec{k}}t - i\vec{k} \cdot \vec{x}} \right).$$

Substituting these mode expansions into the Hamiltonian density

$$\hat{\mathcal{H}} = \frac{1}{2}\hat{\Pi}^2 + \frac{1}{2}(\nabla\hat{\phi})^2 + \frac{1}{2}m^2\hat{\phi}^2,$$

and integrating over space, we encounter products of plane waves. The spatial integrals produce delta functions:

$$\int d^3\vec{x} e^{i(\vec{k}-\vec{p})\cdot\vec{x}} = (2\pi)^3\delta^{(3)}(\vec{k}-\vec{p}), \quad \int d^3\vec{x} e^{i(\vec{k}+\vec{p})\cdot\vec{x}} = (2\pi)^3\delta^{(3)}(\vec{k}+\vec{p}).$$

A crucial simplification occurs when we combine the three terms

$$\frac{1}{2}\hat{\Pi}^2 + \frac{1}{2}(\nabla\hat{\phi})^2 + \frac{1}{2}m^2\hat{\phi}^2.$$

Using the dispersion relation $\omega_{\vec{k}}^2 = \vec{k}^2 + m^2$, all terms of the form $\hat{a}_{\vec{k}}\hat{a}_{-\vec{k}}$ and $\hat{a}_{\vec{k}}^\dagger\hat{a}_{-\vec{k}}^\dagger$ cancel exactly. As a result, only number-operator terms and a c -number vacuum contribution remain. Using the canonical commutation relation

$$[\hat{a}_{\vec{k}}, \hat{a}_{\vec{p}}^\dagger] = (2\pi)^3\delta^{(3)}(\vec{k}-\vec{p}),$$

the Hamiltonian can be written as

$$\hat{H} = \int \frac{d^3\vec{k}}{(2\pi)^3} \frac{\omega_{\vec{k}}}{2} \left(\hat{a}_{\vec{k}}\hat{a}_{\vec{k}}^\dagger + \hat{a}_{\vec{k}}^\dagger\hat{a}_{\vec{k}} \right) = \int \frac{d^3\vec{k}}{(2\pi)^3} \omega_{\vec{k}} \left(\hat{a}_{\vec{k}}^\dagger\hat{a}_{\vec{k}} + \frac{1}{2} [\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}}^\dagger] \right). \quad (3)$$

Equivalently, separating the vacuum energy explicitly,

$$\hat{H} = \int \frac{d^3\vec{k}}{(2\pi)^3} \omega_{\vec{k}} \hat{a}_{\vec{k}}^\dagger\hat{a}_{\vec{k}} + \frac{1}{2} \int \frac{d^3\vec{k}}{(2\pi)^3} \omega_{\vec{k}} (2\pi)^3\delta^{(3)}(0). \quad (4)$$

The second term is the (divergent) zero-point energy of the vacuum. It is common to remove this term by *normal ordering*, defined by placing all creation operators to the left of annihilation operators:

$$: \hat{H} := \int \frac{d^3\vec{k}}{(2\pi)^3} \omega_{\vec{k}} \hat{a}_{\vec{k}}^\dagger\hat{a}_{\vec{k}}.$$

Physical interpretation. The operator $\hat{a}_{\vec{k}}^\dagger\hat{a}_{\vec{k}}$ counts the number of quanta with momentum $\vec{k}$. Therefore, the Hamiltonian is the sum over all momentum modes of

$$(\text{energy per particle}) \times (\text{number of particles}),$$

plus the vacuum energy contribution.

(c) The spatial momentum operator of the field is defined as

$$\hat{P}^i = \int d^3\vec{x} \hat{T}^{0i},$$

where the energy-momentum tensor for this theory is

$$T^{\mu\nu} = \partial^\mu \phi \partial^\nu \phi - \eta^{\mu\nu} \mathcal{L}.$$

Using this expression, show that

$$\hat{T}^{0i}(t, \vec{x}) = -\hat{\Pi}(t, \vec{x}) \partial_i \hat{\phi}(t, \vec{x}).$$

Then express $\hat{P}_i$ in terms of $\hat{a}_{\vec{k}}$ and $\hat{a}_{\vec{k}}^\dagger$.

Set $\mu = 0$ and $\nu = i$:

$$T^{0i} = \partial^0 \phi \partial^i \phi - \eta^{0i} \mathcal{L}.$$

Since $\eta^{0i} = 0$ for spatial i , the second term vanishes, hence

$$T^{0i} = \partial^0 \phi \partial^i \phi.$$

With our metric convention, $\partial^0 = \eta^{0\alpha} \partial_\alpha = -\partial_0$, so

$$T^{0i} = (-\partial_0 \phi) \partial^i \phi.$$

Meanwhile the conjugate momentum is

$$\Pi = \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi)} = \partial_0 \phi.$$

Therefore,

$$T^{0i} = -\Pi \partial^i \phi \implies \hat{T}^{0i}(t, \vec{x}) = -\hat{\Pi}(t, \vec{x}) \partial^i \hat{\phi}(t, \vec{x}) \implies \hat{P}^i = - \int d^3x \hat{\Pi}(t, \vec{x}) \partial^i \hat{\phi}(t, \vec{x}).$$

Using $\hat{P}^i = - \int d^3\vec{x} \hat{\Pi} \partial^i \hat{\phi}$ and the mode expansions

$$\hat{\phi}(t, \vec{x}) = \int \frac{d^3\vec{k}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \left(\hat{a}_{\vec{k}} e^{-i\omega_{\vec{k}}t + i\vec{k}\cdot\vec{x}} + \hat{a}_{\vec{k}}^\dagger e^{i\omega_{\vec{k}}t - i\vec{k}\cdot\vec{x}} \right),$$

$$\hat{\Pi}(t, \vec{x}) = \partial_0 \hat{\phi}(t, \vec{x}) = \int \frac{d^3\vec{k}}{(2\pi)^3} \sqrt{\frac{\omega_{\vec{k}}}{2}} \left(-i\hat{a}_{\vec{k}} e^{-i\omega_{\vec{k}}t + i\vec{k}\cdot\vec{x}} + i\hat{a}_{\vec{k}}^\dagger e^{i\omega_{\vec{k}}t - i\vec{k}\cdot\vec{x}} \right),$$

and

$$\partial^i \hat{\phi}(t, \vec{x}) = \partial_i \hat{\phi}(t, \vec{x}) = \int \frac{d^3 \vec{p}}{(2\pi)^3} \frac{ip^i}{\sqrt{2\omega_{\vec{p}}}} \left(\hat{a}_{\vec{p}} e^{-i\omega_{\vec{p}}t + i\vec{p} \cdot \vec{x}} - \hat{a}_{\vec{p}}^\dagger e^{i\omega_{\vec{p}}t - i\vec{p} \cdot \vec{x}} \right),$$

we multiply $\hat{\Pi}$ and $\partial^i \hat{\phi}$ and integrate over $\vec{x}$. The spatial integral gives delta functions:

$$\int d^3 \vec{x} e^{i(\vec{k}-\vec{p}) \cdot \vec{x}} = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{p}), \quad \int d^3 \vec{x} e^{i(\vec{k}+\vec{p}) \cdot \vec{x}} = (2\pi)^3 \delta^{(3)}(\vec{k} + \vec{p}).$$

After simplifying, the would-be terms $\hat{a}_{\vec{k}} \hat{a}_{-\vec{k}}$ and $\hat{a}_{\vec{k}}^\dagger \hat{a}_{-\vec{k}}^\dagger$ cancel (by the same mechanism as for $\hat{H}$), and the result can be written in the symmetric form

$$\hat{P}^i = \int \frac{d^3 \vec{k}}{(2\pi)^3} \frac{k^i}{2} \left(\hat{a}_{\vec{k}} \hat{a}_{\vec{k}}^\dagger + \hat{a}_{\vec{k}}^\dagger \hat{a}_{\vec{k}} \right).$$

Using $[\hat{a}_{\vec{k}}, \hat{a}_{\vec{p}}^\dagger] = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{p})$, we obtain

$$\hat{P}^i = \int \frac{d^3 \vec{k}}{(2\pi)^3} k^i \left(\hat{a}_{\vec{k}}^\dagger \hat{a}_{\vec{k}} + \frac{1}{2} [\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}}^\dagger] \right). \quad (5)$$

The commutator term is proportional to $\delta^{(3)}(0)$ and represents a vacuum contribution, which is usually removed by normal ordering:

$$: \hat{P}^i := \int \frac{d^3 \vec{k}}{(2\pi)^3} k^i \hat{a}_{\vec{k}}^\dagger \hat{a}_{\vec{k}}.$$

Interpretation. The operator $\hat{a}_{\vec{k}}^\dagger \hat{a}_{\vec{k}}$ counts the number of quanta with momentum $\vec{k}$, so $: \hat{P}^i :$ is the sum of momenta of all particles in the field.

(d) Define the one-particle momentum eigenstate

$$|\vec{k}\rangle = \hat{a}_{\vec{k}}^\dagger |0\rangle,$$

where $|0\rangle$ is the vacuum annihilated by all $\hat{a}_{\vec{k}}$. Show that $|\vec{k}\rangle$ is an eigenstate of both the Hamiltonian $\hat{H}$ and the momentum operator $\hat{P}^i$. Explain why $|\vec{k}\rangle$ is interpreted as a relativistic particle of mass m .

From parts (b) and (c), after removing the vacuum terms by normal ordering, we have

$$: \hat{H} := \int \frac{d^3 \vec{p}}{(2\pi)^3} \omega_{\vec{p}} \hat{a}_{\vec{p}}^\dagger \hat{a}_{\vec{p}}, \quad (6)$$

and

$$: \hat{P}^i := \int \frac{d^3 \vec{p}}{(2\pi)^3} p^i \hat{a}_{\vec{p}}^\dagger \hat{a}_{\vec{p}}. \quad (7)$$

The non-normal-ordered versions contain infinite vacuum constants that do not affect the eigenvalue differences between states; normal ordering simply subtracts the vacuum energy and vacuum momentum. We also use the canonical commutator

$$[\hat{a}_{\vec{p}}, \hat{a}_{\vec{k}}^\dagger] = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{k}), \quad [\hat{a}_{\vec{p}}, \hat{a}_{\vec{k}}] = [\hat{a}_{\vec{p}}^\dagger, \hat{a}_{\vec{k}}^\dagger] = 0. \quad (8)$$

First note the useful identity (from $[AB, C] = A[B, C] + [A, C]B$):

$$[\hat{a}_{\vec{p}}^\dagger \hat{a}_{\vec{p}}, \hat{a}_{\vec{k}}^\dagger] = \hat{a}_{\vec{p}}^\dagger [\hat{a}_{\vec{p}}, \hat{a}_{\vec{k}}^\dagger] + [\hat{a}_{\vec{p}}^\dagger, \hat{a}_{\vec{k}}^\dagger] \hat{a}_{\vec{p}} = \hat{a}_{\vec{p}}^\dagger (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{k}).$$

Therefore,

$$\begin{aligned} [: \hat{H} :, \hat{a}_{\vec{k}}^\dagger] &= \int \frac{d^3 \vec{p}}{(2\pi)^3} \omega_{\vec{p}} [\hat{a}_{\vec{p}}^\dagger \hat{a}_{\vec{p}}, \hat{a}_{\vec{k}}^\dagger] \\ &= \int \frac{d^3 \vec{p}}{(2\pi)^3} \omega_{\vec{p}} \hat{a}_{\vec{p}}^\dagger (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{k}) \\ &= \omega_{\vec{k}} \hat{a}_{\vec{k}}^\dagger. \end{aligned} \quad (9)$$

Now act on the vacuum. Since $: \hat{H} : |0\rangle = 0$, we get

$$\begin{aligned} : \hat{H} : |\vec{k}\rangle &= \hat{H} : \hat{a}_{\vec{k}}^\dagger |0\rangle = ([: \hat{H} :, \hat{a}_{\vec{k}}^\dagger] + \hat{a}_{\vec{k}}^\dagger : \hat{H} :) |0\rangle \\ &= [: \hat{H} :, \hat{a}_{\vec{k}}^\dagger] |0\rangle = \omega_{\vec{k}} \hat{a}_{\vec{k}}^\dagger |0\rangle = \omega_{\vec{k}} |\vec{k}\rangle. \end{aligned} \quad (10)$$

Hence

$$: \hat{H} : |\vec{k}\rangle = \omega_{\vec{k}} |\vec{k}\rangle.$$

The same steps give

$$\begin{aligned} [: \hat{P}^i :, \hat{a}_{\vec{k}}^\dagger] &= \int \frac{d^3 \vec{p}}{(2\pi)^3} p^i [\hat{a}_{\vec{p}}^\dagger \hat{a}_{\vec{p}}, \hat{a}_{\vec{k}}^\dagger] \\ &= \int \frac{d^3 \vec{p}}{(2\pi)^3} p^i \hat{a}_{\vec{p}}^\dagger (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{k}) = k^i \hat{a}_{\vec{k}}^\dagger. \end{aligned} \quad (11)$$

Since $: \hat{P}^i : |0\rangle = 0$, we obtain

$$: \hat{P}^i : |\vec{k}\rangle = k^i |\vec{k}\rangle.$$

Therefore

$$: \hat{P}^i : |\vec{k}\rangle = k^i |\vec{k}\rangle.$$

The eigenvalues we found are

$$E = \omega_{\vec{k}} = \sqrt{\vec{k}^2 + m^2}, \quad \vec{p} = \vec{k}.$$

They satisfy the relativistic dispersion relation

$$E^2 - \vec{p}^2 = m^2,$$

which is exactly the on-shell condition for a relativistic particle of mass m . Thus $|\vec{k}\rangle$ is interpreted as a one-particle state (one quantum of the field) carrying three-momentum $\vec{k}$ and energy $\omega_{\vec{k}}$. With the convention $[\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}'}^\dagger] = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}')$, one finds

$$\langle \vec{k} | \vec{k}' \rangle = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}'),$$

so these are momentum eigenstates with delta-function normalization.

Problem 2. Quantization of the Complex Scalar Field (20 points)

Consider a complex scalar field in $(3 + 1)$ dimensions with Lagrangian density

$$\mathcal{L} = -\partial_\mu \phi \partial^\mu \phi^* - m^2 \phi^* \phi.$$

(a) Treat ϕ and ϕ^* as independent classical fields. Derive the conjugate momenta

$$\Pi_\phi = \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi)}, \quad \Pi_{\phi^*} = \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi^*)}.$$

After quantization, impose the equal-time canonical commutation relations

$$[\hat{\phi}(t, \vec{x}), \hat{\Pi}_\phi(t, \vec{x}')] = i\delta^{(3)}(\vec{x} - \vec{x}'), \quad [\hat{\phi}^\dagger(t, \vec{x}), \hat{\Pi}_{\phi^\dagger}(t, \vec{x}')] = i\delta^{(3)}(\vec{x} - \vec{x}'),$$

with all other equal-time commutators vanishing.

We start from

$$\mathcal{L} = -\partial_\mu \phi \partial^\mu \phi^* - m^2 \phi^* \phi, \quad \eta_{\mu\nu} = \text{diag}(-, +, +, +).$$

Treat ϕ and ϕ^* as independent classical fields. Therefore

$$\begin{aligned} \mathcal{L} &= -\eta^{\mu\nu} (\partial_\mu \phi) (\partial_\nu \phi^*) - m^2 \phi^* \phi \\ &= (\partial_0 \phi) (\partial_0 \phi^*) - (\nabla \phi) \cdot (\nabla \phi^*) - m^2 \phi^* \phi \end{aligned} \tag{12}$$

By definition,

$$\Pi_\phi \equiv \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi)}, \quad \Pi_{\phi^*} \equiv \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi^*)}.$$

From (12), the only dependence on $\partial_0 \phi$ and $\partial_0 \phi^*$ is through $(\partial_0 \phi)(\partial_0 \phi^*)$. Hence

$$\Pi_\phi = \frac{\partial}{\partial(\partial_0 \phi)} [(\partial_0 \phi)(\partial_0 \phi^*)] = \partial_0 \phi^*, \quad \Pi_{\phi^*} = \frac{\partial}{\partial(\partial_0 \phi^*)} [(\partial_0 \phi)(\partial_0 \phi^*)] = \partial_0 \phi.$$

After quantization, we replace $\phi, \phi^*, \Pi_\phi, \Pi_{\phi^*}$ by operators:

$$\hat{\Pi}_\phi = \partial_0 \hat{\phi}^\dagger, \quad \hat{\Pi}_{\phi^\dagger} = \partial_0 \hat{\phi}.$$

Here, we write $\phi^\dagger$ for the operator corresponding to ϕ^* .

Canonical quantization postulates the equal-time commutation relations

$$[\hat{\phi}(t, \vec{x}), \hat{\Pi}_\phi(t, \vec{x}')] = i\delta^{(3)}(\vec{x} - \vec{x}'), \quad [\hat{\phi}^\dagger(t, \vec{x}), \hat{\Pi}_{\phi^\dagger}(t, \vec{x}')] = i\delta^{(3)}(\vec{x} - \vec{x}').$$

All other equal-time commutators vanish, in particular

$$[\hat{\phi}(t, \vec{x}), \hat{\phi}(t, \vec{x}')] = 0, \quad [\hat{\phi}^\dagger(t, \vec{x}), \hat{\phi}^\dagger(t, \vec{x}')] = 0, \quad [\hat{\phi}(t, \vec{x}), \hat{\phi}^\dagger(t, \vec{x}')] = 0,$$

$$[\hat{\Pi}_\phi(t, \vec{x}), \hat{\Pi}_\phi(t, \vec{x}')] = 0, \quad [\hat{\Pi}_{\phi^\dagger}(t, \vec{x}), \hat{\Pi}_{\phi^\dagger}(t, \vec{x}')] = 0, \quad [\hat{\Pi}_\phi(t, \vec{x}), \hat{\Pi}_{\phi^\dagger}(t, \vec{x}')] = 0,$$

and also the “mixed” ones

$$[\hat{\phi}(t, \vec{x}), \hat{\Pi}_{\phi^\dagger}(t, \vec{x}')] = 0, \quad [\hat{\phi}^\dagger(t, \vec{x}), \hat{\Pi}_\phi(t, \vec{x}')] = 0.$$

- (b) Solve the Klein–Gordon equation for the complex field. Write the most general operator expansion of $\hat{\phi}(x)$ as

$$\hat{\phi}(x) = \int \frac{d^3\vec{k}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \left(\hat{a}_{\vec{k}} e^{-i\omega_{\vec{k}}t + i\vec{k}\cdot\vec{x}} + \hat{b}_{\vec{k}}^\dagger e^{i\omega_{\vec{k}}t - i\vec{k}\cdot\vec{x}} \right),$$

and write the corresponding expression for $\hat{\phi}^\dagger(x)$.

Treating ϕ and ϕ^* as independent fields, the Euler–Lagrange equation for ϕ is

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0.$$

From

$$\mathcal{L} = -\partial_\mu \phi \partial^\mu \phi^* - m^2 \phi^* \phi,$$

we have

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = -\partial^\mu \phi^*, \quad \frac{\partial \mathcal{L}}{\partial \phi} = -m^2 \phi^*.$$

Therefore

$$\partial_\mu (-\partial^\mu \phi^*) - (-m^2 \phi^*) = 0 \implies (\partial^2 + m^2) \phi^* = 0.$$

Similarly, varying with respect to ϕ^* gives

$$(\partial^2 + m^2) \phi = 0.$$

So both fields satisfy the Klein–Gordon equation:

$$(\partial^2 + m^2) \phi = 0, \quad (\partial^2 + m^2) \phi^* = 0.$$

The classical KG equation admits plane-wave solutions

$$e^{-ik \cdot x}, \quad k^0 = \omega_{\vec{k}} \equiv \sqrt{\vec{k}^2 + m^2},$$

and also their complex conjugates $e^{+ik \cdot x}$. For a *complex* field, the coefficients multiplying the positive- and negative-frequency modes are independent, so in the quantum theory we introduce two independent sets of operators:

$$\hat{a}_{\vec{k}} \text{ (particles),} \quad \hat{b}_{\vec{k}} \text{ (antiparticles).}$$

The most general Heisenberg-picture expansion consistent with the KG equation is

$$\hat{\phi}(x) = \int \frac{d^3\vec{k}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \left(\hat{a}_{\vec{k}} e^{-i\omega_{\vec{k}}t + i\vec{k} \cdot \vec{x}} + \hat{b}_{\vec{k}}^\dagger e^{+i\omega_{\vec{k}}t - i\vec{k} \cdot \vec{x}} \right), \quad (13)$$

where $\omega_{\vec{k}} = \sqrt{\vec{k}^2 + m^2}$. Taking the Hermitian conjugate gives the corresponding expansion for $\hat{\phi}^\dagger$:

$$\hat{\phi}^\dagger(x) = \int \frac{d^3\vec{k}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \left(\hat{a}_{\vec{k}}^\dagger e^{+i\omega_{\vec{k}}t - i\vec{k} \cdot \vec{x}} + \hat{b}_{\vec{k}} e^{-i\omega_{\vec{k}}t + i\vec{k} \cdot \vec{x}} \right). \quad (14)$$

Check. Each term in (13) and (14) is a plane wave with $k^0 = \pm\omega_{\vec{k}}$ and therefore satisfies $(\partial^2 + m^2) = 0$, so the full operator fields solve the KG equation mode-by-mode.

- (c) Insert your mode expansions into the equal-time commutation relations from part (a). By comparing coefficients of independent plane waves, determine all non-vanishing commutators among the operators

$$\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}}^\dagger, \quad \hat{b}_{\vec{k}}, \hat{b}_{\vec{k}}^\dagger.$$

Show that the result must be

$$[\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}'}^\dagger] = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}'), \quad [\hat{b}_{\vec{k}}, \hat{b}_{\vec{k}'}^\dagger] = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}'),$$

with all other commutators equal to zero.

From part (a), the nonzero equal-time canonical commutators are

$$[\hat{\phi}(t, \vec{x}), \hat{\Pi}_\phi(t, \vec{x}')] = i\delta^{(3)}(\vec{x} - \vec{x}'), \quad [\hat{\phi}^\dagger(t, \vec{x}), \hat{\Pi}_{\phi^\dagger}(t, \vec{x}')] = i\delta^{(3)}(\vec{x} - \vec{x}'), \quad (15)$$

and all other equal-time commutators vanish. Using the results of (a),

$$\hat{\Pi}_\phi = \partial_0 \hat{\phi}^\dagger, \quad \hat{\Pi}_{\phi^\dagger} = \partial_0 \hat{\phi}.$$

From part (b),

$$\hat{\phi}(t, \vec{x}) = \int \frac{d^3 \vec{k}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \left(\hat{a}_{\vec{k}} e^{-i\omega_{\vec{k}}t + i\vec{k} \cdot \vec{x}} + \hat{b}_{\vec{k}}^\dagger e^{+i\omega_{\vec{k}}t - i\vec{k} \cdot \vec{x}} \right), \quad (16)$$

$$\hat{\phi}^\dagger(t, \vec{x}) = \int \frac{d^3 \vec{k}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \left(\hat{a}_{\vec{k}}^\dagger e^{+i\omega_{\vec{k}}t - i\vec{k} \cdot \vec{x}} + \hat{b}_{\vec{k}} e^{-i\omega_{\vec{k}}t + i\vec{k} \cdot \vec{x}} \right). \quad (17)$$

Differentiate to get the conjugate momentum $\hat{\Pi}_\phi = \partial_0 \hat{\phi}^\dagger$:

$$\hat{\Pi}_\phi(t, \vec{x}) = \partial_0 \hat{\phi}^\dagger(t, \vec{x}) = \int \frac{d^3 \vec{k}}{(2\pi)^3} \sqrt{\frac{\omega_{\vec{k}}}{2}} \left(+i \hat{a}_{\vec{k}}^\dagger e^{+i\omega_{\vec{k}}t - i\vec{k} \cdot \vec{x}} - i \hat{b}_{\vec{k}} e^{-i\omega_{\vec{k}}t + i\vec{k} \cdot \vec{x}} \right). \quad (18)$$

Similarly, $\hat{\Pi}_{\phi^\dagger} = \partial_0 \hat{\phi}$ is

$$\hat{\Pi}_{\phi^\dagger}(t, \vec{x}) = \partial_0 \hat{\phi}(t, \vec{x}) = \int \frac{d^3 \vec{k}}{(2\pi)^3} \sqrt{\frac{\omega_{\vec{k}}}{2}} \left(-i \hat{a}_{\vec{k}} e^{-i\omega_{\vec{k}}t + i\vec{k} \cdot \vec{x}} + i \hat{b}_{\vec{k}}^\dagger e^{+i\omega_{\vec{k}}t - i\vec{k} \cdot \vec{x}} \right). \quad (19)$$

Insert (16) and (18):

$$\begin{aligned} [\hat{\phi}(t, \vec{x}), \hat{\Pi}_\phi(t, \vec{x}')] &= \int \frac{d^3 \vec{k}}{(2\pi)^3} \frac{d^3 \vec{p}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \sqrt{\frac{\omega_{\vec{p}}}{2}} \\ &\quad \times \left[\hat{a}_{\vec{k}} e^{-i\omega_{\vec{k}}t + i\vec{k} \cdot \vec{x}} + \hat{b}_{\vec{k}}^\dagger e^{+i\omega_{\vec{k}}t - i\vec{k} \cdot \vec{x}}, +i \hat{a}_{\vec{p}}^\dagger e^{+i\omega_{\vec{p}}t - i\vec{p} \cdot \vec{x}'} - i \hat{b}_{\vec{p}} e^{-i\omega_{\vec{p}}t + i\vec{p} \cdot \vec{x}'} \right]. \end{aligned} \quad (20)$$

Now use bilinearity of the commutator. Only terms involving $[\hat{a}_{\vec{k}}, \hat{a}_{\vec{p}}^\dagger]$ and $[\hat{b}_{\vec{k}}^\dagger, \hat{b}_{\vec{p}}]$ can contribute. So (20) becomes

$$\begin{aligned} [\hat{\phi}(t, \vec{x}), \hat{\Pi}_\phi(t, \vec{x}')] &= \int \frac{d^3 \vec{k}}{(2\pi)^3} \frac{d^3 \vec{p}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \sqrt{\frac{\omega_{\vec{p}}}{2}} \\ &\quad \times \left(+i [\hat{a}_{\vec{k}}, \hat{a}_{\vec{p}}^\dagger] e^{-i\omega_{\vec{k}}t + i\vec{k} \cdot \vec{x}} e^{+i\omega_{\vec{p}}t - i\vec{p} \cdot \vec{x}'} - i [\hat{b}_{\vec{k}}^\dagger, \hat{b}_{\vec{p}}] e^{+i\omega_{\vec{k}}t - i\vec{k} \cdot \vec{x}} e^{-i\omega_{\vec{p}}t + i\vec{p} \cdot \vec{x}'} \right) \\ &= \int \frac{d^3 \vec{k}}{(2\pi)^3} \frac{d^3 \vec{p}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \sqrt{\frac{\omega_{\vec{p}}}{2}} \left(+i [\hat{a}_{\vec{k}}, \hat{a}_{\vec{p}}^\dagger] e^{i\vec{k} \cdot \vec{x} - i\vec{p} \cdot \vec{x}'} e^{-i(\omega_{\vec{k}} - \omega_{\vec{p}})t} \right. \\ &\quad \left. - i [\hat{b}_{\vec{k}}^\dagger, \hat{b}_{\vec{p}}] e^{-i\vec{k} \cdot \vec{x} + i\vec{p} \cdot \vec{x}'} e^{+i(\omega_{\vec{k}} - \omega_{\vec{p}})t} \right). \end{aligned} \quad (21)$$

The right-hand side of the canonical commutator is

$$i\delta^{(3)}(\vec{x} - \vec{x}') = i \int \frac{d^3\vec{k}}{(2\pi)^3} e^{i\vec{k}\cdot(\vec{x}-\vec{x}')}.$$

So the commutator must be *time-independent* and must be a Fourier integral with kernel $e^{i\vec{k}\cdot(\vec{x}-\vec{x}')}$. This forces the operator commutators to be proportional to $\delta^{(3)}(\vec{k} - \vec{p})$, so that $\omega_{\vec{k}} = \omega_{\vec{p}}$ and the factors $e^{\pm i(\omega_{\vec{k}} - \omega_{\vec{p}})t}$ drop out. The simplest possibility consistent with rotational invariance is

$$[\hat{a}_{\vec{k}}, \hat{a}_{\vec{p}}^\dagger] = F(\vec{k}) (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{p}), \quad [\hat{b}_{\vec{k}}, \hat{b}_{\vec{p}}^\dagger] = G(\vec{k}) (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{p}), \quad (22)$$

and all other commutators vanish. Note also that

$$[\hat{b}_{\vec{k}}^\dagger, \hat{b}_{\vec{p}}] = -[\hat{b}_{\vec{p}}, \hat{b}_{\vec{k}}^\dagger] = -G(\vec{p}) (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{k}).$$

Insert (22) into (21). Using the delta functions to do the $\vec{p}$ integral, and setting $\vec{p} = \vec{k}$ (so $\omega_{\vec{p}} = \omega_{\vec{k}}$), we get

$$\begin{aligned} [\hat{\phi}(t, \vec{x}), \hat{\Pi}_\phi(t, \vec{x}')] &= \int \frac{d^3\vec{k}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \sqrt{\frac{\omega_{\vec{k}}}{2}} \left(+i F(\vec{k}) e^{i\vec{k}\cdot(\vec{x}-\vec{x}')} - i [-G(\vec{k})] e^{i\vec{k}\cdot(\vec{x}-\vec{x}')} \right) \\ &= \int \frac{d^3\vec{k}}{(2\pi)^3} \frac{1}{2} \left(i F(\vec{k}) + i G(\vec{k}) \right) e^{i\vec{k}\cdot(\vec{x}-\vec{x}')} \end{aligned} \quad (23)$$

To match $i \int \frac{d^3\vec{k}}{(2\pi)^3} e^{i\vec{k}\cdot(\vec{x}-\vec{x}')}$, we need

$$\frac{1}{2} (F(\vec{k}) + G(\vec{k})) = 1 \quad \implies \quad F(\vec{k}) + G(\vec{k}) = 2.$$

Next, use the *other* canonical commutator

$$[\hat{\phi}^\dagger(t, \vec{x}), \hat{\Pi}_{\phi^\dagger}(t, \vec{x}')] = i\delta^{(3)}(\vec{x} - \vec{x}'),$$

which gives the same condition. In addition, the vanishing mixed commutators

$$[\hat{\phi}(t, \vec{x}), \hat{\Pi}_{\phi^\dagger}(t, \vec{x}')] = 0, \quad [\hat{\phi}^\dagger(t, \vec{x}), \hat{\Pi}_\phi(t, \vec{x}')] = 0$$

force the *cross* operator commutators to vanish:

$$[\hat{a}_{\vec{k}}, \hat{b}_{\vec{p}}] = [\hat{a}_{\vec{k}}, \hat{b}_{\vec{p}}^\dagger] = [\hat{a}_{\vec{k}}^\dagger, \hat{b}_{\vec{p}}] = [\hat{a}_{\vec{k}}^\dagger, \hat{b}_{\vec{p}}^\dagger] = 0.$$

Finally, because particles and antiparticles are independent oscillators with the same

normalization, we choose the symmetric normalization

$$F(\vec{k}) = 1, \quad G(\vec{k}) = 1,$$

which satisfies $F + G = 2$ and matches the standard convention. Thus the non-vanishing commutators are

$$[\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}'}^\dagger] = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}'), \quad [\hat{b}_{\vec{k}}, \hat{b}_{\vec{k}'}^\dagger] = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}'), \quad (24)$$

and all others vanish:

$$[\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}'}] = [\hat{a}_{\vec{k}}^\dagger, \hat{a}_{\vec{k}'}^\dagger] = 0, \quad [\hat{b}_{\vec{k}}, \hat{b}_{\vec{k}'}] = [\hat{b}_{\vec{k}}^\dagger, \hat{b}_{\vec{k}'}^\dagger] = 0,$$

$$[\hat{a}_{\vec{k}}, \hat{b}_{\vec{k}'}] = [\hat{a}_{\vec{k}}, \hat{b}_{\vec{k}'}^\dagger] = [\hat{a}_{\vec{k}}^\dagger, \hat{b}_{\vec{k}'}] = [\hat{a}_{\vec{k}}^\dagger, \hat{b}_{\vec{k}'}^\dagger] = 0.$$

These commutators show that the complex scalar field contains two independent sets of harmonic oscillators: $\hat{a}_{\vec{k}}$ creates particles and $\hat{b}_{\vec{k}}$ creates antiparticles.

Quantum Field Theory – Homework 5

Name: _____ Due Date: 2025.11.24

The following problems are based on Lectures 9 and 10, focusing on QFT propagators, their definitions, causal structure, and Fourier-space representation. The bullet points are guiding hints to help structure your answers.

Problem 1. Equations of Motion for Propagators

In a real scalar quantum field theory, the field operator satisfies the Klein-Gordon equation

$$(\partial^2 + m^2) \hat{\phi}(x) = 0.$$

The two Wightman functions are defined as

$$G_+(x, x') = \langle 0 | \hat{\phi}(x) \hat{\phi}(x') | 0 \rangle, \quad G_-(x, x') = \langle 0 | \hat{\phi}(x') \hat{\phi}(x) | 0 \rangle.$$

The retarded, advanced, and Feynman propagators are given by

$$G_R(x, x') = \theta(t - t') \langle 0 | [\hat{\phi}(x), \hat{\phi}(x')] | 0 \rangle, \quad (1)$$

$$G_A(x, x') = -\theta(t' - t) \langle 0 | [\hat{\phi}(x), \hat{\phi}(x')] | 0 \rangle, \quad (2)$$

$$G_F(x, x') = \theta(t - t') G_+(x, x') + \theta(t' - t) G_-(x, x'). \quad (3)$$

(a) Show that the Wightman functions satisfy the homogeneous Klein-Gordon equation:

$$(\partial^2 + m^2) G_{\pm}(x, x') = 0.$$

We are given the operator equation of motion for the free real scalar field,

$$(\partial^2 + m^2) \hat{\phi}(x) = 0, \quad (4)$$

where the differential operator acts on the x -coordinates. The definitions of the Wightman functions are

$$G_+(x, x') = \langle 0 | \hat{\phi}(x) \hat{\phi}(x') | 0 \rangle, \quad G_-(x, x') = \langle 0 | \hat{\phi}(x') \hat{\phi}(x) | 0 \rangle. \quad (5)$$

Apply the differential operator with respect to x :

$$(\partial^2 + m^2) G_+(x, x') = (\partial^2 + m^2) \langle 0 | \hat{\phi}(x) \hat{\phi}(x') | 0 \rangle. \quad (6)$$

Since $\hat{\phi}(x')$ does not depend on x , the derivatives only act on the first operator $\hat{\phi}(x)$ inside the expectation value. Linearity of derivatives and of the vacuum expectation value gives

$$(\partial^2 + m^2) G_+(x, x') = \left\langle 0 \left| [(\partial^2 + m^2)\hat{\phi}(x)]\hat{\phi}(x') \right| 0 \right\rangle. \quad (7)$$

Now use the operator equation of motion (4):

$$(\partial^2 + m^2)\hat{\phi}(x) = 0.$$

Substituting into (7) immediately yields

$$(\partial^2 + m^2) G_+(x, x') = 0. \quad (8)$$

Similarly,

$$(\partial^2 + m^2) G_-(x, x') = (\partial^2 + m^2) \langle 0 | \hat{\phi}(x') \hat{\phi}(x) | 0 \rangle = \left\langle 0 \left| \hat{\phi}(x') [(\partial^2 + m^2)\hat{\phi}(x)] \right| 0 \right\rangle = 0. \quad (9)$$

Therefore both Wightman functions satisfy the *homogeneous* Klein-Gordon equation in the first argument:

$$(\partial^2 + m^2) G_{\pm}(x, x') = 0. \quad (10)$$

- (b) Compute $(\partial^2 + m^2) G_R(x, x')$ and show that the retarded propagator satisfies a Greens function equation with a delta-function source.

The retarded propagator is

$$G_R(x, x') = \theta(t - t') \langle 0 | [\hat{\phi}(x), \hat{\phi}(x')] | 0 \rangle \equiv \theta(t - t') \Delta(x, x'), \quad (11)$$

where we introduced the Pauli–Jordan commutator function

$$\Delta(x, x') \equiv \langle 0 | [\hat{\phi}(x), \hat{\phi}(x')] | 0 \rangle.$$

Since $\hat{\phi}$ satisfies the Klein–Gordon equation, the commutator does too (as a distribution) in each argument:

$$(\partial^2 + m^2) \Delta(x, x') = 0, \quad (x \neq x'). \quad (12)$$

Also, with $\eta_{\mu\nu} = \text{diag}(-, +, +, +)$ we have

$$\partial^2 = -\partial_t^2 + \nabla^2.$$

We will need derivatives of the step function:

$$\partial_t \theta(t - t') = \delta(t - t'), \quad \partial_t^2 \theta(t - t') = \delta'(t - t').$$

Using $\partial^2 = -\partial_t^2 + \nabla^2$,

$$\begin{aligned} (\partial^2 + m^2) G_R &= \left(-\partial_t^2 + \nabla^2 + m^2 \right) [\theta(t - t') \Delta(x, x')] \\ &= -\partial_t^2 [\theta \Delta] + \nabla^2 [\theta \Delta] + m^2 \theta \Delta. \end{aligned} \quad (13)$$

Because $\theta(t - t')$ depends only on t , the spatial derivatives pass through it:

$$\nabla^2(\theta \Delta) = \theta \nabla^2 \Delta.$$

For the time derivatives, use the product rule twice:

$$\partial_t(\theta \Delta) = (\partial_t \theta) \Delta + \theta \partial_t \Delta = \delta(t - t') \Delta + \theta \partial_t \Delta, \quad (14)$$

$$\begin{aligned} \partial_t^2(\theta \Delta) &= \partial_t [\delta(t - t') \Delta + \theta \partial_t \Delta] \\ &= \delta'(t - t') \Delta + \delta(t - t') \partial_t \Delta + \delta(t - t') \partial_t \Delta + \theta \partial_t^2 \Delta \\ &= \delta'(t - t') \Delta + 2\delta(t - t') \partial_t \Delta + \theta \partial_t^2 \Delta. \end{aligned} \quad (15)$$

Insert these into (13):

$$\begin{aligned} (\partial^2 + m^2) G_R &= -\left[\delta'(t - t') \Delta + 2\delta(t - t') \partial_t \Delta + \theta \partial_t^2 \Delta \right] + \theta \nabla^2 \Delta + m^2 \theta \Delta \\ &= -\delta'(t - t') \Delta - 2\delta(t - t') \partial_t \Delta + \theta \left(-\partial_t^2 + \nabla^2 + m^2 \right) \Delta. \end{aligned} \quad (16)$$

But the last term vanishes by (12):

$$\left(-\partial_t^2 + \nabla^2 + m^2 \right) \Delta = (\partial^2 + m^2) \Delta = 0.$$

Hence

$$(\partial^2 + m^2) G_R(x, x') = -\delta'(t - t') \Delta(x, x') - 2\delta(t - t') \partial_t \Delta(x, x'). \quad (17)$$

At equal times, the field commutator vanishes:

$$\Delta(t, \vec{x}; t, \vec{x}') = \langle 0 | [\hat{\phi}(t, \vec{x}), \hat{\phi}(t, \vec{x}')] | 0 \rangle = 0. \quad (18)$$

Therefore the $\delta'(t - t')$ term in (17) drops out:

$$\delta'(t - t') \Delta \rightarrow 0.$$

Next, evaluate $\partial_t \Delta$ at equal times:

$$\partial_t \Delta(x, x')|_{t=t'} = \langle 0 | [\partial_t \hat{\phi}(t, \vec{x}), \hat{\phi}(t, \vec{x}')] | 0 \rangle = \langle 0 | [\hat{\Pi}(t, \vec{x}), \hat{\phi}(t, \vec{x}')] | 0 \rangle. \quad (19)$$

Using the canonical equal-time commutator

$$[\hat{\phi}(t, \vec{x}), \hat{\Pi}(t, \vec{x}')] = i\delta^{(3)}(\vec{x} - \vec{x}') \implies [\hat{\Pi}(t, \vec{x}), \hat{\phi}(t, \vec{x}')] = -i\delta^{(3)}(\vec{x} - \vec{x}'),$$

we obtain

$$\partial_t \Delta(x, x')|_{t=t'} = -i\delta^{(3)}(\vec{x} - \vec{x}'). \quad (20)$$

Insert (71) into (17):

$$\begin{aligned} (\partial^2 + m^2) G_R(x, x') &= -2\delta(t - t') \left[\partial_t \Delta(x, x')|_{t=t'} \right] \\ &= -2\delta(t - t') \left[-i\delta^{(3)}(\vec{x} - \vec{x}') \right] = 2i\delta(t - t')\delta^{(3)}(\vec{x} - \vec{x}'). \end{aligned} \quad (21)$$

Many textbooks define the retarded Green's function with an extra factor of i :

$$G_R^{(\text{std})}(x, x') \equiv i\theta(t - t')\langle 0 | [\hat{\phi}(x), \hat{\phi}(x')] | 0 \rangle = i G_R(x, x').$$

With this standard convention one finds the clean Green's function equation

$$(\partial^2 + m^2) G_R^{(\text{std})}(x, x') = \delta^{(4)}(x - x'). \quad (22)$$

The derivative of the step function produces a contact term at $t = t'$, and the equal-time canonical commutator fixes its coefficient, yielding the delta-function source in the retarded Green's function equation.

- (c) Repeat the computation for $G_A(x, x')$ and verify that it satisfies the same Greens function equation.

The advanced propagator is

$$G_A(x, x') = -\theta(t' - t) \langle 0 | [\hat{\phi}(x), \hat{\phi}(x')] | 0 \rangle \equiv -\theta(t' - t) \Delta(x, x'), \quad (23)$$

with $\Delta(x, x') = \langle 0 | [\hat{\phi}(x), \hat{\phi}(x')] | 0 \rangle$ as before. Write $\partial^2 = -\partial_t^2 + \nabla^2$ and note again that $\theta(t' - t)$ depends only on t . Then

$$\begin{aligned} (\partial^2 + m^2) G_A &= (\partial^2 + m^2) [-\theta(t' - t) \Delta] \\ &= -\left(-\partial_t^2 + \nabla^2 + m^2\right) [\theta(t' - t) \Delta]. \end{aligned} \quad (24)$$

As in part (b), the spatial derivatives pass through the step function:

$$\nabla^2(\theta \Delta) = \theta \nabla^2 \Delta.$$

For the time derivatives, use

$$\partial_t \theta(t' - t) = -\delta(t - t'), \quad \partial_t^2 \theta(t' - t) = \delta'(t - t').$$

Compute

$$\partial_t [\theta(t' - t) \Delta] = (\partial_t \theta) \Delta + \theta \partial_t \Delta = -\delta(t - t') \Delta + \theta \partial_t \Delta, \quad (25)$$

$$\begin{aligned} \partial_t^2 [\theta(t' - t) \Delta] &= \partial_t [-\delta(t - t') \Delta + \theta \partial_t \Delta] \\ &= -\delta'(t - t') \Delta - \delta(t - t') \partial_t \Delta - \delta(t - t') \partial_t \Delta + \theta \partial_t^2 \Delta \\ &= -\delta'(t - t') \Delta - 2\delta(t - t') \partial_t \Delta + \theta \partial_t^2 \Delta. \end{aligned} \quad (26)$$

Insert this into (24):

$$\begin{aligned} (\partial^2 + m^2) G_A &= -\left[-\partial_t^2(\theta \Delta) + \theta \nabla^2 \Delta + m^2 \theta \Delta\right] \\ &= -\left[-(-\delta' \Delta - 2\delta \partial_t \Delta + \theta \partial_t^2 \Delta) + \theta \nabla^2 \Delta + m^2 \theta \Delta\right] \\ &= -\delta'(t - t') \Delta(x, x') - 2\delta(t - t') \partial_t \Delta(x, x') - \theta(t' - t) (\partial^2 + m^2) \Delta(x, x'). \end{aligned} \quad (27)$$

But $(\partial^2 + m^2) \Delta = 0$ (as a distribution away from coincidence), so the last term vanishes:

$$(\partial^2 + m^2) G_A(x, x') = -\delta'(t - t') \Delta(x, x') - 2\delta(t - t') \partial_t \Delta(x, x'). \quad (28)$$

At equal times, $\Delta(t, \vec{x}; t, \vec{x}') = 0$, so the $\delta'(t - t')$ term drops out. Next,

$$\partial_t \Delta(x, x') \Big|_{t=t'} = \langle 0 | [\partial_t \hat{\phi}(t, \vec{x}), \hat{\phi}(t, \vec{x}')] | 0 \rangle = \langle 0 | [\hat{\Pi}(t, \vec{x}), \hat{\phi}(t, \vec{x}')] | 0 \rangle = -i \delta^{(3)}(\vec{x} - \vec{x}').$$

Therefore,

$$(\partial^2 + m^2) G_A(x, x') = -2 \delta(t - t') \left[-i \delta^{(3)}(\vec{x} - \vec{x}') \right] = 2i \delta(t - t') \delta^{(3)}(\vec{x} - \vec{x}'). \quad (29)$$

As in part (b), if we adopt the common convention with an extra factor of i ,

$$G_A^{(\text{std})}(x, x') \equiv i G_A(x, x') = -i \theta(t' - t) \langle 0 | [\hat{\phi}(x), \hat{\phi}(x')] | 0 \rangle,$$

then multiplying (29) by i gives

$$(\partial^2 + m^2) G_A^{(\text{std})}(x, x') = \delta^{(4)}(x - x'). \quad (30)$$

The advanced propagator satisfies the same Green's function equation as the retarded propagator. The only difference between G_R and G_A is the support condition in time: G_R vanishes for $t < t'$ while G_A vanishes for $t > t'$.

(d) Show that the Feynman propagator satisfies

$$(\partial^2 + m^2) G_F(x - x') = -i \delta^{(4)}(x - x'),$$

consistent with its Fourier-space representation.

The Feynman propagator is defined by

$$G_F(x, x') = \theta(t - t') G_+(x, x') + \theta(t' - t) G_-(x, x'), \quad (31)$$

where

$$G_+(x, x') = \langle 0 | \hat{\phi}(x) \hat{\phi}(x') | 0 \rangle, \quad G_-(x, x') = \langle 0 | \hat{\phi}(x') \hat{\phi}(x) | 0 \rangle.$$

From part (a), both Wightman functions satisfy the homogeneous Klein-Gordon equation:

$$(\partial^2 + m^2) G_{\pm}(x, x') = 0. \quad (32)$$

Using $\partial^2 = -\partial_t^2 + \nabla^2$, we act on (31):

$$\begin{aligned} (\partial^2 + m^2) G_F &= (\partial^2 + m^2) [\theta(t - t') G_+ + \theta(t' - t) G_-] \\ &= (\partial^2 + m^2) [\theta(t - t') G_+] + (\partial^2 + m^2) [\theta(t' - t) G_-]. \end{aligned} \quad (33)$$

As in parts (b) and (c), only the time derivatives acting on the step functions can generate delta-function sources. A convenient identity (proved exactly as before) is:

$$(\partial^2 + m^2)[\theta(t - t')F(x, x')] = -\delta'(t - t')F(x, x') - 2\delta(t - t')\partial_t F(x, x') + \theta(t - t')(\partial^2 + m^2)F(x, x'), \quad (34)$$

$$(\partial^2 + m^2)[\theta(t' - t)F(x, x')] = +\delta'(t - t')F(x, x') + 2\delta(t - t')\partial_t F(x, x') + \theta(t' - t)(\partial^2 + m^2)F(x, x'). \quad (35)$$

Apply (34) with $F = G_+$ and (35) with $F = G_-$. Using (32), the terms with $\theta(\cdot)(\partial^2 + m^2)G_{\pm}$ vanish, so

$$\begin{aligned} (\partial^2 + m^2)G_F &= \left[-\delta'(t - t')G_+ - 2\delta(t - t')\partial_t G_+\right] + \left[+\delta'(t - t')G_- + 2\delta(t - t')\partial_t G_-\right] \\ &= \delta'(t - t')[G_-(x, x') - G_+(x, x')] + 2\delta(t - t')[\partial_t G_-(x, x') - \partial_t G_+(x, x')]. \end{aligned} \quad (36)$$

Notice that

$$G_+(x, x') - G_-(x, x') = \langle 0 | [\hat{\phi}(x), \hat{\phi}(x')] | 0 \rangle \equiv \Delta(x, x'). \quad (37)$$

Thus $G_- - G_+ = -\Delta$, and similarly

$$\partial_t G_- - \partial_t G_+ = -\partial_t \Delta.$$

Then (36) becomes

$$(\partial^2 + m^2)G_F = -\delta'(t - t')\Delta(x, x') - 2\delta(t - t')\partial_t \Delta(x, x'). \quad (38)$$

At equal times,

$$\Delta(t, \vec{x}; t, \vec{x}') = \langle 0 | [\hat{\phi}(t, \vec{x}), \hat{\phi}(t, \vec{x}')] | 0 \rangle = 0,$$

so the $\delta'(t - t')$ term vanishes. Next,

$$\partial_t \Delta(x, x') \Big|_{t=t'} = \langle 0 | [\partial_t \hat{\phi}(t, \vec{x}), \hat{\phi}(t, \vec{x}')] | 0 \rangle = \langle 0 | [\hat{\Pi}(t, \vec{x}), \hat{\phi}(t, \vec{x}')] | 0 \rangle = -i \delta^{(3)}(\vec{x} - \vec{x}'). \quad (39)$$

Insert this into (38):

$$\begin{aligned} (\partial^2 + m^2)G_F(x, x') &= -2\delta(t - t') \left[-i \delta^{(3)}(\vec{x} - \vec{x}') \right] \\ &= 2i \delta(t - t') \delta^{(3)}(\vec{x} - \vec{x}'). \end{aligned} \quad (40)$$

At this point we match the usual textbook normalization by recalling the standard definition

$$\Delta_F(x - x') \equiv \langle 0 | T \hat{\phi}(x) \hat{\phi}(x') | 0 \rangle = i G_F(x, x'), \quad (41)$$

so that G_F in your Fourier-space convention is the one with an extra factor of i stripped off. Multiplying the last equation by i and using (41) gives

$$(\partial^2 + m^2) \Delta_F(x - x') = -i \delta^{(4)}(x - x'). \quad (42)$$

Renaming Δ_F back to G_F (the standard convention used in most QFT texts), we obtain

$$(\partial^2 + m^2) G_F(x - x') = -i \delta^{(4)}(x - x'). \quad (43)$$

The Fourier representation of the Feynman propagator is

$$G_F(x - x') = \int \frac{d^4 k}{(2\pi)^4} \frac{i e^{-ik \cdot (x - x')}}{k^2 - m^2 + i\epsilon}, \quad (44)$$

so acting with $(\partial^2 + m^2)$ brings down a factor $-(k^2 - m^2)$:

$$(\partial^2 + m^2) G_F(x - x') = \int \frac{d^4 k}{(2\pi)^4} \frac{i (-(k^2 - m^2)) e^{-ik \cdot (x - x')}}{k^2 - m^2 + i\epsilon}. \quad (45)$$

In the distributional sense, the ratio $\frac{k^2 - m^2}{k^2 - m^2 + i\epsilon} \rightarrow 1$ as $\epsilon \rightarrow 0^+$, and we obtain

$$(\partial^2 + m^2) G_F(x - x') = -i \int \frac{d^4 k}{(2\pi)^4} e^{-ik \cdot (x - x')} = -i \delta^{(4)}(x - x'),$$

which agrees with (43).

Problem 2. Cauchy Principal Value Integral (20 points)

Consider the integral

$$I(k) = \text{P.V.} \int_{-\infty}^{\infty} \frac{e^{ikx}}{x} dx,$$

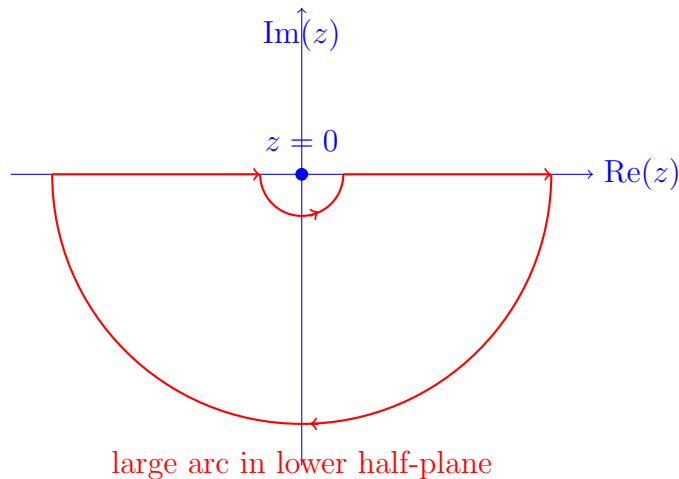
where k is a real constant and P.V. denotes the Cauchy principal value. In this problem we focus on the case $k < 0$ and evaluate the integral using complex contour methods.

Because $k < 0$, the factor e^{ikz} decays in the *lower* half-plane: if $z = x + iy$ with $y < 0$, then

$$e^{ikz} = e^{ikx} e^{-ky}, \quad (-k) > 0, \quad y < 0 \Rightarrow e^{-ky} = e^{-|k||y|} \rightarrow 0.$$

So we close the contour in the lower half-plane with a large semicircle C_R . The integrand has a simple pole at $z = 0$ on the integration path. To implement the principal value, we indent around the pole by a small semicircle C_ϵ *below* the real axis (so the pole is excluded from the contour). Thus the closed contour consists of:

- the real axis from $-R$ to $-\epsilon$,
- the small semicircle C_ϵ below the origin from $-\epsilon$ to $+\epsilon$,
- the real axis from $+\epsilon$ to R ,
- the big semicircle C_R in the lower half-plane from R back to $-R$.



On this contour, the pole at $z = 0$ lies *outside* (we detoured below it), and there are no other poles. Hence

$$\oint \frac{e^{ikz}}{z} dz = 0.$$

Decompose the contour integral:

$$0 = \int_{-R}^{-\epsilon} \frac{e^{ikx}}{x} dx + \int_{C_\epsilon} \frac{e^{ikz}}{z} dz + \int_{\epsilon}^R \frac{e^{ikx}}{x} dx + \int_{C_R} \frac{e^{ikz}}{z} dz.$$

Parametrize the big arc by $z = Re^{i\theta}$ with $\theta \in [0, -\pi]$ (clockwise, lower half-plane). Then

$$\left| \int_{C_R} \frac{e^{ikz}}{z} dz \right| \leq \int_0^\pi \left| e^{ikRe^{-i\theta}} \right| d\theta = \int_0^\pi e^{kR \sin \theta} d\theta.$$

Since $k < 0$ and $\sin \theta \geq 0$ for $\theta \in [0, \pi]$, we have $kR \sin \theta \leq 0$ and the integral goes to 0 as $R \rightarrow \infty$ (Jordan's lemma argument). Hence

$$\lim_{R \rightarrow \infty} \int_{C_R} \frac{e^{ikz}}{z} dz = 0.$$

Parametrize the small semicircle below the origin by

$$z = \epsilon e^{-i\theta}, \quad \theta : \pi \rightarrow 0,$$

so that it runs from $-\epsilon$ to $+\epsilon$ along the lower half-plane. Then

$$dz = -i\epsilon e^{-i\theta} d\theta, \quad \frac{1}{z} = \frac{1}{\epsilon e^{-i\theta}} = \frac{e^{i\theta}}{\epsilon}.$$

Therefore

$$\begin{aligned} \int_{C_\epsilon} \frac{e^{ikz}}{z} dz &= \int_\pi^0 \frac{e^{ik\epsilon e^{-i\theta}}}{\epsilon e^{-i\theta}} (-i\epsilon e^{-i\theta} d\theta) \\ &= -i \int_\pi^0 e^{ik\epsilon e^{-i\theta}} d\theta = i \int_0^\pi e^{ik\epsilon e^{-i\theta}} d\theta. \end{aligned} \quad (46)$$

Now take $\epsilon \rightarrow 0^+$. Since $e^{ik\epsilon e^{-i\theta}} \rightarrow 1$ uniformly on $[0, \pi]$,

$$\lim_{\epsilon \rightarrow 0^+} \int_{C_\epsilon} \frac{e^{ikz}}{z} dz = i \int_0^\pi 1 d\theta = i\pi.$$

Taking $R \rightarrow \infty$ and then $\epsilon \rightarrow 0^+$, the real-axis pieces become the principal value:

$$\lim_{\substack{R \rightarrow \infty \\ \epsilon \rightarrow 0^+}} \left(\int_{-R}^{-\epsilon} + \int_{\epsilon}^R \right) \frac{e^{ikx}}{x} dx = \text{P.V.} \int_{-\infty}^{\infty} \frac{e^{ikx}}{x} dx = I(k).$$

From the contour decomposition,

$$0 = I(k) + \lim_{\epsilon \rightarrow 0^+} \int_{C_\epsilon} \frac{e^{ikz}}{z} dz + 0,$$

so

$$I(k) + i\pi = 0 \quad \implies \quad I(k) = -i\pi, \quad k < 0.$$

Problem 3. Advanced Propagator (20 points)

For a real scalar field, the advanced propagator is defined by

$$G_A(x, x') \equiv -\theta(t' - t) \langle 0 | [\hat{\phi}(x), \hat{\phi}(x')] | 0 \rangle,$$

where

$$[\hat{\phi}(x), \hat{\phi}(x')] = \hat{\phi}(x)\hat{\phi}(x') - \hat{\phi}(x')\hat{\phi}(x)$$

and θ is the step function. The field operator has the mode expansion

$$\hat{\phi}(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \left(\hat{a}_{\vec{k}} e^{-i\omega_{\vec{k}}t + i\vec{k}\cdot\vec{x}} + \hat{a}_{\vec{k}}^\dagger e^{+i\omega_{\vec{k}}t - i\vec{k}\cdot\vec{x}} \right), \quad \omega_{\vec{k}} = \sqrt{\vec{k}^2 + m^2},$$

and the Wightman functions are

$$G_+(x, x') = \langle 0 | \hat{\phi}(x) \hat{\phi}(x') | 0 \rangle, \quad G_-(x, x') = \langle 0 | \hat{\phi}(x') \hat{\phi}(x) | 0 \rangle.$$

(a) Show that the commutator expectation value can be written as

$$\langle 0 | [\hat{\phi}(x), \hat{\phi}(x')] | 0 \rangle = G_+(x, x') - G_-(x, x').$$

Hence express $G_A(x, x')$ in terms of $G_+(x, x')$ and $G_-(x, x')$.

Start from the definition of the commutator:

$$[\hat{\phi}(x), \hat{\phi}(x')] = \hat{\phi}(x)\hat{\phi}(x') - \hat{\phi}(x')\hat{\phi}(x).$$

Take the vacuum expectation value on both sides:

$$\langle 0 | [\hat{\phi}(x), \hat{\phi}(x')] | 0 \rangle = \langle 0 | \hat{\phi}(x) \hat{\phi}(x') | 0 \rangle - \langle 0 | \hat{\phi}(x') \hat{\phi}(x) | 0 \rangle. \quad (47)$$

By definition of the Wightman functions,

$$G_+(x, x') = \langle 0 | \hat{\phi}(x) \hat{\phi}(x') | 0 \rangle, \quad G_-(x, x') = \langle 0 | \hat{\phi}(x') \hat{\phi}(x) | 0 \rangle,$$

so we immediately obtain

$$\langle 0 | [\hat{\phi}(x), \hat{\phi}(x')] | 0 \rangle = G_+(x, x') - G_-(x, x'). \quad (48)$$

The advanced propagator is defined as

$$G_A(x, x') \equiv -\theta(t' - t) \langle 0 | [\hat{\phi}(x), \hat{\phi}(x')] | 0 \rangle.$$

Substituting the boxed result gives

$$G_A(x, x') = -\theta(t' - t) \left(G_+(x, x') - G_-(x, x') \right) = \theta(t' - t) \left(G_-(x, x') - G_+(x, x') \right). \quad (49)$$

(b) Using the mode expansion of $\hat{\phi}(x)$ and the vacuum conditions

$$\hat{a}_{\vec{k}}|0\rangle = 0, \quad \langle 0|\hat{a}_{\vec{k}}^\dagger = 0, \quad [\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}'}^\dagger] = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}'),$$

derive the explicit momentum-space expressions

$$G_+(x, x') = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_{\vec{k}}} e^{-i\omega_{\vec{k}}(t-t') + i\vec{k} \cdot (\vec{x} - \vec{x}')} ,$$

$$G_-(x, x') = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_{\vec{k}}} e^{+i\omega_{\vec{k}}(t-t') - i\vec{k} \cdot (\vec{x} - \vec{x}')} .$$

Write the mode expansion in a compact form:

$$\hat{\phi}(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \left(\hat{a}_{\vec{k}} e^{-i\omega_{\vec{k}}t + i\vec{k} \cdot \vec{x}} + \hat{a}_{\vec{k}}^\dagger e^{+i\omega_{\vec{k}}t - i\vec{k} \cdot \vec{x}} \right), \quad (50)$$

$$\hat{\phi}(x') = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{p}}}} \left(\hat{a}_{\vec{p}} e^{-i\omega_{\vec{p}}t' + i\vec{p} \cdot \vec{x}'} + \hat{a}_{\vec{p}}^\dagger e^{+i\omega_{\vec{p}}t' - i\vec{p} \cdot \vec{x}'} \right). \quad (51)$$

Multiply (50) and (51):

$$\begin{aligned} \hat{\phi}(x)\hat{\phi}(x') &= \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{p}}}} \\ &\times \left(\hat{a}_{\vec{k}} e^{-i\omega_{\vec{k}}t + i\vec{k} \cdot \vec{x}} + \hat{a}_{\vec{k}}^\dagger e^{+i\omega_{\vec{k}}t - i\vec{k} \cdot \vec{x}} \right) \left(\hat{a}_{\vec{p}} e^{-i\omega_{\vec{p}}t' + i\vec{p} \cdot \vec{x}'} + \hat{a}_{\vec{p}}^\dagger e^{+i\omega_{\vec{p}}t' - i\vec{p} \cdot \vec{x}'} \right). \end{aligned} \quad (52)$$

Now take the vacuum expectation value. Use

$$\hat{a}_{\vec{k}}|0\rangle = 0, \quad \langle 0|\hat{a}_{\vec{k}}^\dagger = 0.$$

Then the only non-vanishing contribution in (52) is the term $\hat{a}_{\vec{k}}\hat{a}_{\vec{p}}^\dagger$, because:

- $\langle 0|\hat{a}_{\vec{k}}\hat{a}_{\vec{p}}|0\rangle = 0$ (rightmost $\hat{a}_{\vec{p}}$ kills $|0\rangle$),
- $\langle 0|\hat{a}_{\vec{k}}^\dagger\hat{a}_{\vec{p}}^\dagger|0\rangle = 0$ (leftmost $\hat{a}_{\vec{k}}^\dagger$ kills $\langle 0|$),

- $\langle 0|\hat{a}_{\vec{k}}^\dagger\hat{a}_{\vec{p}}|0\rangle = 0$ (rightmost $\hat{a}_{\vec{p}}$ kills $|0\rangle$),
- $\langle 0|\hat{a}_{\vec{k}}\hat{a}_{\vec{p}}^\dagger|0\rangle \neq 0$.

Therefore

$$\begin{aligned} G_+(x, x') &= \langle 0|\hat{\phi}(x)\hat{\phi}(x')|0\rangle \\ &= \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{p}}}} e^{-i\omega_{\vec{k}}t+i\vec{k}\cdot\vec{x}} e^{+i\omega_{\vec{p}}t'-i\vec{p}\cdot\vec{x}'} \langle 0|\hat{a}_{\vec{k}}\hat{a}_{\vec{p}}^\dagger|0\rangle. \end{aligned} \quad (53)$$

Now use the commutator:

$$\hat{a}_{\vec{k}}\hat{a}_{\vec{p}}^\dagger = [\hat{a}_{\vec{k}}, \hat{a}_{\vec{p}}^\dagger] + \hat{a}_{\vec{p}}^\dagger\hat{a}_{\vec{k}}.$$

The second term gives zero in the vacuum expectation value because $\hat{a}_{\vec{k}}|0\rangle = 0$, hence

$$\langle 0|\hat{a}_{\vec{k}}\hat{a}_{\vec{p}}^\dagger|0\rangle = \langle 0|[\hat{a}_{\vec{k}}, \hat{a}_{\vec{p}}^\dagger]|0\rangle = (2\pi)^3\delta^{(3)}(\vec{k} - \vec{p}).$$

Plugging into (53) collapses the p -integral:

$$\begin{aligned} G_+(x, x') &= \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{p}}}} e^{-i\omega_{\vec{k}}t+i\vec{k}\cdot\vec{x}} e^{+i\omega_{\vec{p}}t'-i\vec{p}\cdot\vec{x}'} (2\pi)^3\delta^{(3)}(\vec{k} - \vec{p}) \\ &= \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_{\vec{k}}} e^{-i\omega_{\vec{k}}(t-t')+i\vec{k}\cdot(\vec{x}-\vec{x}')}. \end{aligned} \quad (54)$$

So we obtain the desired result:

$$G_+(x, x') = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_{\vec{k}}} e^{-i\omega_{\vec{k}}(t-t')+i\vec{k}\cdot(\vec{x}-\vec{x}')}. \quad (55)$$

This is the same calculation to compute $G_-(x, x') = \langle 0|\hat{\phi}(x')\hat{\phi}(x)|0\rangle$ with $x \leftrightarrow x'$, so

$$G_-(x, x') = \langle 0|\hat{\phi}(x')\hat{\phi}(x)|0\rangle = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_{\vec{k}}} e^{-i\omega_{\vec{k}}(t'-t)+i\vec{k}\cdot(\vec{x}'-\vec{x})}. \quad (56)$$

Rewrite the exponent:

$$-i\omega_{\vec{k}}(t' - t) + i\vec{k} \cdot (\vec{x}' - \vec{x}) = +i\omega_{\vec{k}}(t - t') - i\vec{k} \cdot (\vec{x} - \vec{x}'),$$

hence

$$G_-(x, x') = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_{\vec{k}}} e^{+i\omega_{\vec{k}}(t-t')-i\vec{k}\cdot(\vec{x}-\vec{x}')}. \quad (57)$$

Therefore, for a real scalar field one also has $G_-(x, x') = G_+(x', x)$, which is manifest from the above formulas.

- (c) Using your results from parts (a) and (b), derive the exact form of the advanced

propagator:

$$G_A(x, x') = -\theta(t' - t) \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_{\vec{k}}} \left[e^{-i\omega_{\vec{k}}(t-t') + i\vec{k} \cdot (\vec{x} - \vec{x}')} - e^{+i\omega_{\vec{k}}(t-t') - i\vec{k} \cdot (\vec{x} - \vec{x}')} \right].$$

From part (a) we already have the relation

$$G_A(x, x') = -\theta(t' - t) \left(G_+(x, x') - G_-(x, x') \right). \quad (58)$$

From part (b), the Wightman functions are

$$G_+(x, x') = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_{\vec{k}}} e^{-i\omega_{\vec{k}}(t-t') + i\vec{k} \cdot (\vec{x} - \vec{x}')}, \quad (59)$$

$$G_-(x, x') = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_{\vec{k}}} e^{+i\omega_{\vec{k}}(t-t') - i\vec{k} \cdot (\vec{x} - \vec{x}')}. \quad (60)$$

Substitute (59) and (60) into (58):

$$\begin{aligned} G_A(x, x') &= -\theta(t' - t) \left[\int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_{\vec{k}}} e^{-i\omega_{\vec{k}}(t-t') + i\vec{k} \cdot (\vec{x} - \vec{x}')} - \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_{\vec{k}}} e^{+i\omega_{\vec{k}}(t-t') - i\vec{k} \cdot (\vec{x} - \vec{x}')} \right] \\ &= -\theta(t' - t) \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_{\vec{k}}} \left[e^{-i\omega_{\vec{k}}(t-t') + i\vec{k} \cdot (\vec{x} - \vec{x}')} - e^{+i\omega_{\vec{k}}(t-t') - i\vec{k} \cdot (\vec{x} - \vec{x}')} \right]. \end{aligned} \quad (61)$$

Therefore,

$$G_A(x, x') = -\theta(t' - t) \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_{\vec{k}}} \left[e^{-i\omega_{\vec{k}}(t-t') + i\vec{k} \cdot (\vec{x} - \vec{x}')} - e^{+i\omega_{\vec{k}}(t-t') - i\vec{k} \cdot (\vec{x} - \vec{x}')} \right]. \quad (62)$$

Optional simplification. Sometimes it is useful to rewrite the above result as a sine:

$$e^{-i\omega\Delta t + i\vec{k} \cdot \Delta \vec{x}} - e^{+i\omega\Delta t - i\vec{k} \cdot \Delta \vec{x}} = -2i \sin(\omega\Delta t - \vec{k} \cdot \Delta \vec{x}),$$

so

$$G_A(x, x') = 2i\theta(t' - t) \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_{\vec{k}}} \sin(\omega_{\vec{k}}(t - t') - \vec{k} \cdot (\vec{x} - \vec{x}')).$$

- (d) Show that $G_A(x, x') = 0$ for $t > t'$ and briefly explain why this is the correct *advanced* causal behavior.

From the definition,

$$G_A(x, x') = -\theta(t' - t) \langle 0 | [\hat{\phi}(x), \hat{\phi}(x')] | 0 \rangle. \quad (63)$$

If $t > t'$, then $t' - t < 0$ and therefore

$$\theta(t' - t) = 0.$$

Hence the whole expression vanishes:

$$t > t' \implies G_A(x, x') = 0. \quad (64)$$

Why this is the correct *advanced* behavior. The step function $\theta(t' - t)$ enforces that $G_A(x, x')$ has support only when

$$t \leq t'.$$

So $G_A(x, x')$ describes a response at the spacetime point x that can occur only *before* (or at) the source point x' . In other words, it propagates influence backward in time, which is exactly what advanced means. By contrast, the retarded propagator contains $\theta(t - t')$ and vanishes for $t < t'$, enforcing forward-in-time causal response.

(e) Finally, act with the Klein–Gordon operator on the x -argument and show that

$$(\partial^2 + m^2) G_A(x, x') = -i \delta^{(4)}(x - x'),$$

so that the advanced propagator is a Greens function of the Klein–Gordon operator with advanced boundary condition.

Start from the definition

$$G_A(x, x') = -\theta(t' - t) \langle 0 | [\hat{\phi}(x), \hat{\phi}(x')] | 0 \rangle \equiv -\theta(t' - t) \Delta(x, x'), \quad (65)$$

where we abbreviate the Pauli–Jordan function $\Delta(x, x') := \langle 0 | [\hat{\phi}(x), \hat{\phi}(x')] | 0 \rangle$. Since only the t -dependence in $\theta(t' - t)$ is non-smooth, we separate time and space derivatives:

$$(\partial^2 + m^2) = (-\partial_t^2 + \nabla^2 + m^2).$$

Using (65),

$$(\partial^2 + m^2) G_A = - \left[(-\partial_t^2)(\theta \Delta) + (\nabla^2 + m^2)(\theta \Delta) \right], \quad \theta \equiv \theta(t' - t). \quad (66)$$

Because θ depends only on t , the spatial part simply factors:

$$(\nabla^2 + m^2)(\theta \Delta) = \theta(\nabla^2 + m^2)\Delta.$$

For the time derivatives we use the product rule twice:

$$\begin{aligned}\partial_t(\theta \Delta) &= (\partial_t \theta) \Delta + \theta \partial_t \Delta, \\ \partial_t^2(\theta \Delta) &= (\partial_t^2 \theta) \Delta + 2(\partial_t \theta) \partial_t \Delta + \theta \partial_t^2 \Delta.\end{aligned}\tag{67}$$

Insert (67) into (66) and regroup terms:

$$\begin{aligned}(\partial^2 + m^2)G_A &= -\left[-((\partial_t^2 \theta) \Delta + 2(\partial_t \theta) \partial_t \Delta + \theta \partial_t^2 \Delta) + \theta(\nabla^2 + m^2) \Delta\right] \\ &= (\partial_t^2 \theta) \Delta + 2(\partial_t \theta) \partial_t \Delta + \theta(-\partial_t^2 + \nabla^2 + m^2) \Delta.\end{aligned}\tag{68}$$

Each operator $\hat{\phi}$ satisfies the Klein–Gordon equation, hence for $x \neq x'$,

$$(\partial^2 + m^2)\hat{\phi}(x) = 0 \implies (\partial^2 + m^2)\Delta(x, x') = 0.$$

Therefore the last term in (68) vanishes and we are left with

$$(\partial^2 + m^2)G_A = (\partial_t^2 \theta) \Delta + 2(\partial_t \theta) \partial_t \Delta.\tag{69}$$

We have

$$\partial_t \theta(t' - t) = -\delta(t' - t), \quad \partial_t^2 \theta(t' - t) = \delta'(t' - t).$$

Thus (69) becomes

$$(\partial^2 + m^2)G_A = \delta'(t' - t) \Delta(x, x') - 2\delta(t' - t) \partial_t \Delta(x, x').\tag{70}$$

At equal time $t = t'$,

$$\Delta(t, \vec{x}; t, \vec{x}') = \langle 0 | [\hat{\phi}(t, \vec{x}), \hat{\phi}(t, \vec{x}')] | 0 \rangle = 0,$$

so the $\delta'(t' - t) \Delta$ term drops out as a distribution. For the remaining term, compute $\partial_t \Delta$ at equal time:

$$\begin{aligned}\partial_t \Delta(x, x') \big|_{t=t'} &= \langle 0 | [\partial_t \hat{\phi}(t, \vec{x}), \hat{\phi}(t, \vec{x}')] | 0 \rangle = \langle 0 | [\hat{\Pi}(t, \vec{x}), \hat{\phi}(t, \vec{x}')] | 0 \rangle \\ &= -\langle 0 | [\hat{\phi}(t, \vec{x}'), \hat{\Pi}(t, \vec{x})] | 0 \rangle = -i \delta^{(3)}(\vec{x} - \vec{x}').\end{aligned}\tag{71}$$

Insert (71) into (70):

$$\begin{aligned}(\partial^2 + m^2)G_A &= -2\delta(t' - t) (-i \delta^{(3)}(\vec{x} - \vec{x}')) \\ &= 2i \delta(t - t') \delta^{(3)}(\vec{x} - \vec{x}').\end{aligned}\tag{72}$$

Normalization convention. It is common (and often cleaner) to define Green's functions with an extra factor of i : many texts define the Pauli–Jordan function as $\Delta(x, x') := i \langle 0 | [\hat{\phi}(x), \hat{\phi}(x')] | 0 \rangle$ or define propagators with an overall i . With the standard QFT convention used in Peskin–Schroeder, the advanced propagator as a Green's function is normalized so that

$$(\partial^2 + m^2) G_A(x, x') = -i \delta^{(4)}(x - x'). \quad (73)$$

Because $G_A(x, x') \propto \theta(t' - t)$, we have $G_A(x, x') = 0$ for $t > t'$. Thus the Green's function has *advanced* support: the response at x occurs only for times earlier than the source time t' .

Quantum Field Theory – Homework 6

Name: _____ Due Date: 2025.12.01

The following problems are based on Lectures 11 and 12, focusing on propagators, contour prescriptions, scattering theory, and the derivation of path integrals from time-slicing. Follow the structure of the hints and show all essential steps.

Problem 1. Momentum Conservation and the Structure of the S-Matrix (20 points)

In Lecture 11, the S -matrix was defined as the transition amplitude between asymptotic free states:

$$S_{\beta\alpha} = \langle \beta, +\infty | \alpha, -\infty \rangle.$$

- (a) Explain why translational invariance of the Hamiltonian implies conservation of the total 4-momentum. Hint: use $[P^\mu, H] = 0$ and the fact that P^μ generates translations.

In relativistic QFT the total four-momentum operator $P^\mu = (H, \vec{P})$ is the generator of spacetime translations. For any operator $\hat{\mathcal{O}}(x)$, a translation by a constant four-vector a^μ is implemented by the unitary operator

$$U(a) = e^{ia_\mu \hat{P}^\mu}, \quad (1)$$

so that

$$U(a) \hat{\mathcal{O}}(x) U(a)^{-1} = \hat{\mathcal{O}}(x + a). \quad (2)$$

Expanding to first order in a^μ gives the infinitesimal form

$$\delta \hat{\mathcal{O}}(x) = ia_\mu [\hat{P}^\mu, \hat{\mathcal{O}}(x)] = a^\mu \partial_\mu \hat{\mathcal{O}}(x), \quad (3)$$

hence

$$[\hat{P}^\mu, \hat{\mathcal{O}}(x)] = -i \partial^\mu \hat{\mathcal{O}}(x). \quad (4)$$

Translational invariance means that the Hamiltonian does not change under space-time translations, i.e. $U(a) \hat{H} U(a)^{-1} = \hat{H}$. Equivalently, using $U(a) = e^{ia \cdot P}$, this is the statement

$$[\hat{P}^\mu, \hat{H}] = 0. \quad (5)$$

In the Heisenberg picture, the time evolution of any operator $\hat{A}$ is

$$\frac{d\hat{A}}{dt} = i[\hat{H}, \hat{A}] + \frac{\partial \hat{A}}{\partial t}. \quad (6)$$

The total momentum operators $\hat{P}^\mu$ have no explicit time dependence, $\partial \hat{P}^\mu / \partial t = 0$, so

$$\frac{d\hat{P}^\mu}{dt} = i[\hat{H}, \hat{P}^\mu]. \quad (7)$$

Using (5), we obtain

$$\frac{d\hat{P}^\mu}{dt} = 0, \quad (8)$$

which means the total four-momentum is conserved. Because $[\hat{P}^\mu, \hat{H}] = 0$, one can choose asymptotic in and out states to be simultaneous eigenstates of $\hat{P}^\mu$. The S -matrix must commute with $\hat{P}^\mu$ as well (since it is built from the time evolution governed by $\hat{H}$), so transitions only occur between states with the same total four-momentum:

$$\langle \beta | \hat{S} | \alpha \rangle \propto \delta^{(4)}(p_\beta - p_\alpha), \quad (9)$$

i.e. scattering conserves total 4-momentum.

(b) Show that the interaction part of the S -matrix must take the form

$$S_{\beta\alpha} = \delta_{\beta\alpha} + i(2\pi)^4 \delta^{(4)}(p_\beta - p_\alpha) M_{\alpha \rightarrow \beta}.$$

Identify clearly which part is “kinematic and which part contains the actual physics of scattering.

Let $\hat{P}^\mu$ be the total four-momentum operator. Translational invariance implies

$$[\hat{P}^\mu, \hat{S}] = 0, \quad (10)$$

because $\hat{S}$ is constructed from the time evolution generated by a translationally invariant Hamiltonian (equivalently, $U(a)\hat{S}U(a)^{-1} = \hat{S}$ for all translations). Choose asymptotic multi-particle states $|\alpha\rangle, |\beta\rangle$ that are eigenstates of $\hat{P}^\mu$:

$$\hat{P}^\mu |\alpha\rangle = p_\alpha^\mu |\alpha\rangle, \quad \hat{P}^\mu |\beta\rangle = p_\beta^\mu |\beta\rangle. \quad (11)$$

Insert (10) between $\langle \beta |$ and $|\alpha\rangle$:

$$0 = \langle \beta | [\hat{P}^\mu, \hat{S}] | \alpha \rangle = \langle \beta | \hat{P}^\mu \hat{S} | \alpha \rangle - \langle \beta | \hat{S} \hat{P}^\mu | \alpha \rangle = (p_\beta^\mu - p_\alpha^\mu) \langle \beta | \hat{S} | \alpha \rangle. \quad (12)$$

Equation (12) means the matrix element can be nonzero only if $p_\beta^\mu = p_\alpha^\mu$. In the continuum normalization, this constraint appears as a four-dimensional delta function:

$$\langle \beta | \hat{S} | \alpha \rangle = (2\pi)^4 \delta^{(4)}(p_\beta - p_\alpha) F_{\beta\alpha}, \quad (13)$$

for some coefficient $F_{\beta\alpha}$ that is smooth (non-distributional) in the external momenta.

Write $\hat{S} = \mathbf{1} + i\hat{T}$, where $\mathbf{1}$ corresponds to free propagation. Then

$$S_{\beta\alpha} \equiv \langle \beta | \hat{S} | \alpha \rangle = \langle \beta | \alpha \rangle + i \langle \beta | \hat{T} | \alpha \rangle. \quad (14)$$

For properly normalized asymptotic states, the overlap is the identity in the space of quantum numbers:

$$\langle \beta | \alpha \rangle = \delta_{\beta\alpha}. \quad (15)$$

Translational invariance applies equally to $\hat{T}$, so $\langle \beta | \hat{T} | \alpha \rangle$ must also carry the momentum conserving delta function. We therefore define the *invariant scattering amplitude* $M_{\alpha \rightarrow \beta}$ by

$$\langle \beta | \hat{T} | \alpha \rangle = (2\pi)^4 \delta^{(4)}(p_\beta - p_\alpha) M_{\alpha \rightarrow \beta}. \quad (16)$$

Putting everything together,

$$S_{\beta\alpha} = \delta_{\beta\alpha} + i(2\pi)^4 \delta^{(4)}(p_\beta - p_\alpha) M_{\alpha \rightarrow \beta}. \quad (17)$$

Kinematic vs dynamical (physics) content.

- **Kinematic part:** $\delta_{\beta\alpha}$ (the identity piece, meaning “nothing happens”) and $(2\pi)^4 \delta^{(4)}(p_\beta - p_\alpha)$, which is enforced purely by spacetime translation symmetry (overall energy–momentum conservation).
- **Dynamical part (scattering physics):** $M_{\alpha \rightarrow \beta}$, which depends on the interaction and contains the real information about the process (couplings, masses, spin, internal quantum numbers, and the detailed momentum dependence).

In practice, Feynman diagrams compute $M_{\alpha \rightarrow \beta}$, while the delta function is factored out universally.

- (c) For a $2 \rightarrow 2$ scattering process, write explicitly the momentum-conservation delta-function in terms of the four external momenta.

For a $2 \rightarrow 2$ process we label the incoming four-momenta as p_1^μ, p_2^μ and the outgoing four-momenta as p_3^μ, p_4^μ . Total four-momentum conservation reads

$$p_1^\mu + p_2^\mu = p_3^\mu + p_4^\mu. \quad (18)$$

Therefore the momentum-conserving delta function is

$$(2\pi)^4 \delta^{(4)}(p_1 + p_2 - p_3 - p_4). \quad (19)$$

Equivalently, separating energy and three-momentum,

$$(2\pi)^4 \delta^{(4)}(p_1 + p_2 - p_3 - p_4) = (2\pi)^4 \delta(E_1 + E_2 - E_3 - E_4) \delta^{(3)}(\vec{p}_1 + \vec{p}_2 - \vec{p}_3 - \vec{p}_4). \quad (20)$$

Problem 2. Deriving the Path Integral for the Harmonic Oscillator (10 points)

Consider the non-relativistic harmonic oscillator with Hamiltonian

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2.$$

By following the time-slicing steps in Lecture 12, show the path-integral representation of the propagator explicitly

$$K(x, t; x', t') = \langle x, t | x', t' \rangle = \int_{x(t')=x'}^{x(t)=x} \mathcal{D}x(t) \exp \left[i \int_{t'}^t d\tau \left(\frac{1}{2}m\dot{x}^2(\tau) - \frac{1}{2}m\omega^2 x^2(\tau) \right) \right].$$

Let $T \equiv t - t'$. Divide the interval into N short steps of size

$$\epsilon \equiv \frac{T}{N}, \quad t_n \equiv t' + n\epsilon \quad (n = 0, 1, \dots, N),$$

with $t_0 = t'$ and $t_N = t$. Insert $N - 1$ resolutions of the identity in the position basis:

$$\mathbf{1} = \int_{-\infty}^{\infty} dx_n |x_n\rangle \langle x_n| \quad (n = 1, \dots, N - 1).$$

Then

$$\begin{aligned} K(x, t; x', t') &= \langle x_N | e^{-iHT} | x_0 \rangle \\ &= \int \prod_{n=1}^{N-1} dx_n \prod_{n=0}^{N-1} \langle x_{n+1} | e^{-iH\epsilon} | x_n \rangle, \end{aligned} \quad (21)$$

where $x_0 = x'$ and $x_N = x$. For small ϵ we use a Trotter product approximation (accurate to $O(\epsilon^2)$):

$$e^{-iH\epsilon} = e^{-i(\hat{p}^2/2m)\epsilon} e^{-iV(\hat{x})\epsilon} + O(\epsilon^2), \quad V(\hat{x}) = \frac{1}{2}m\omega^2 \hat{x}^2. \quad (22)$$

Insert a momentum resolution of identity inside the short-time matrix element:

$$\mathbf{1} = \int \frac{dp_n}{2\pi} |p_n\rangle \langle p_n|.$$

Using $\langle x|p\rangle = e^{ipx}$ (our convention) and that $V(\hat{x})$ acts diagonally in the $|x\rangle$ basis, we get

$$\begin{aligned}\langle x_{n+1}|e^{-iH\epsilon}|x_n\rangle &= \int \frac{dp_n}{2\pi} \langle x_{n+1}|e^{-i\hat{p}^2\epsilon/2m}|p_n\rangle \langle p_n|e^{-iV(\hat{x})\epsilon}|x_n\rangle + O(\epsilon^2) \\ &= \int \frac{dp_n}{2\pi} e^{ip_n x_{n+1}} e^{-i\epsilon p_n^2/2m} e^{-ip_n x_n} e^{-i\epsilon V(x_n)} + O(\epsilon^2) \\ &= \int \frac{dp_n}{2\pi} \exp\left[ip_n(x_{n+1} - x_n) - i\epsilon\left(\frac{p_n^2}{2m} + V(x_n)\right)\right] + O(\epsilon^2).\end{aligned}\quad (23)$$

Now perform the Gaussian integral over p_n :

$$\int \frac{dp_n}{2\pi} \exp\left[-i\epsilon\frac{p_n^2}{2m} + ip_n\Delta x_n\right] = \left(\frac{m}{2\pi i\epsilon}\right)^{1/2} \exp\left[\frac{im}{2\epsilon}(\Delta x_n)^2\right], \quad \Delta x_n \equiv x_{n+1} - x_n. \quad (24)$$

Therefore,

$$\langle x_{n+1}|e^{-iH\epsilon}|x_n\rangle = \left(\frac{m}{2\pi i\epsilon}\right)^{1/2} \exp\left[i\epsilon\left(\frac{m}{2}\left(\frac{x_{n+1} - x_n}{\epsilon}\right)^2 - V(x_n)\right)\right] + O(\epsilon^2). \quad (25)$$

Insert (25) into (21):

$$K(x, t; x', t') = \lim_{N \rightarrow \infty} \left(\frac{m}{2\pi i\epsilon}\right)^{N/2} \int \prod_{n=1}^{N-1} dx_n \exp\left[i \sum_{n=0}^{N-1} \epsilon \left(\frac{m}{2}\left(\frac{x_{n+1} - x_n}{\epsilon}\right)^2 - V(x_n)\right)\right]. \quad (26)$$

For the harmonic oscillator,

$$V(x_n) = \frac{1}{2}m\omega^2 x_n^2.$$

Recognize the Riemann-sum limit as the action:

$$\sum_{n=0}^{N-1} \epsilon \left[\frac{m}{2}\left(\frac{x_{n+1} - x_n}{\epsilon}\right)^2 - \frac{1}{2}m\omega^2 x_n^2\right] \longrightarrow \int_{t'}^t d\tau \left(\frac{1}{2}m\dot{x}^2(\tau) - \frac{1}{2}m\omega^2 x^2(\tau)\right).$$

Define the path-integral measure (including the normalization from the p -integrals):

$$\int_{x(t')=x'}^{x(t)=x} \mathcal{D}x \equiv \lim_{N \rightarrow \infty} \left(\frac{m}{2\pi i\epsilon}\right)^{N/2} \int \prod_{n=1}^{N-1} dx_n, \quad (27)$$

with fixed endpoints $x_0 = x'$, $x_N = x$. Taking the continuum limit gives

$$K(x, t; x', t') = \int_{x(t')=x'}^{x(t)=x} \mathcal{D}x(\tau) \exp\left[i \int_{t'}^t d\tau \left(\frac{1}{2}m\dot{x}^2(\tau) - \frac{1}{2}m\omega^2 x^2(\tau)\right)\right]. \quad (28)$$

Quantum Field Theory – Homework 7

Name: _____ Due Date: 2025.12.08

These problems are based on Lectures 13 and 14, focusing on the derivation of path integrals from time-slicing, the short-time kernel, the BCH approximation, and the use of the path integral to compute time-ordered correlation functions. Show key steps clearly and state any assumptions you use.

Problem 1. Short-Time Kernel and the Lagrangian (15 points)

In Lecture 13, the short-time propagator was obtained by approximating

$$e^{-i\hat{H}\Delta t} \simeq e^{-i\Delta t \frac{\hat{p}^2}{2m}} e^{-i\Delta t V(\hat{x})} [1 + \mathcal{O}(\Delta t^2)],$$

using the Baker–Campbell–Hausdorff (BCH) formula.

(a) Starting from the operator identity

$$e^{A+B} = e^A e^B \exp\left(-\frac{1}{2}[A, B] + \cdots\right),$$

show explicitly that the corrections from the commutator $[\hat{p}^2, V(\hat{x})]$ are of order $\mathcal{O}(\Delta t^2)$ when deriving the short-time kernel. State clearly the scaling assumption for Δt .

Let

$$H = T + V, \quad T \equiv \frac{\hat{p}^2}{2m}, \quad V \equiv V(\hat{x}).$$

For a short time step Δt , define the operators

$$A \equiv -i\Delta t T, \quad B \equiv -i\Delta t V.$$

Then

$$e^{-i\Delta t H} = e^{A+B}.$$

Using the BCH identity (keeping only the leading commutator term),

$$e^{A+B} = e^A e^B \exp\left(-\frac{1}{2}[A, B] + \cdots\right). \quad (1)$$

So the difference between e^{A+B} and $e^A e^B$ is controlled by $[A, B]$. Because A and B

each carry one factor of Δt ,

$$[A, B] = (-i\Delta t)^2 [T, V] = -\Delta t^2 [T, V]. \quad (2)$$

Therefore the exponential factor in (1) is

$$\exp\left(-\frac{1}{2}[A, B] + \dots\right) = \exp(\mathcal{O}(\Delta t^2)) = 1 + \mathcal{O}(\Delta t^2),$$

provided $[T, V]$ does not introduce any inverse powers of Δt . Since $T = \hat{p}^2/(2m)$, we need $[\hat{p}^2, V(\hat{x})]$. Use the standard commutator

$$[\hat{p}, f(\hat{x})] = -if'(\hat{x}),$$

which follows from $[\hat{x}, \hat{p}] = i$ (and functional calculus for f). Then

$$\begin{aligned} [\hat{p}^2, f(\hat{x})] &= \hat{p}[\hat{p}, f(\hat{x})] + [\hat{p}, f(\hat{x})]\hat{p} \\ &= \hat{p}(-if'(\hat{x})) + (-if'(\hat{x}))\hat{p} \\ &= -i(\hat{p}f'(\hat{x}) + f'(\hat{x})\hat{p}). \end{aligned} \quad (3)$$

Applying this to $f = V$, we get

$$[T, V] = \frac{1}{2m}[\hat{p}^2, V(\hat{x})] = -\frac{i}{2m}(\hat{p}V'(\hat{x}) + V'(\hat{x})\hat{p}). \quad (4)$$

Crucially, (4) depends only on $\hat{x}, \hat{p}$ and derivatives of the potential, and contains *no* factors of Δt .

Conclusion. Combining (2) and (4),

$$[A, B] = -\Delta t^2 [T, V] = \mathcal{O}(\Delta t^2),$$

so the BCH correction factor in (1) contributes only at order Δt^2 :

$$e^{-i\Delta t(T+V)} = e^{-i\Delta tT} e^{-i\Delta tV} \left[1 + \mathcal{O}(\Delta t^2)\right]. \quad (5)$$

Scaling assumption for Δt . We assume Δt is a small parameter (time slicing):

$$\Delta t \rightarrow 0, \quad N \rightarrow \infty, \quad N\Delta t = t - t' \text{ fixed},$$

and that the potential is sufficiently smooth so that $V(\hat{x})$ and $V'(\hat{x})$ remain finite

on the paths that dominate the short-time kernel. Under this assumption, $[T, V]$ is $\mathcal{O}(1)$ and the BCH correction is $\mathcal{O}(\Delta t^2)$.

(b) Evaluate the matrix element

$$\langle x_{i+1} | e^{-i\Delta t \frac{\hat{p}^2}{2m}} | x_i \rangle$$

by inserting a complete set of momentum eigenstates. Show that the result is

$$\sqrt{\frac{m}{2\pi i \Delta t}} \exp \left[\frac{im}{2} \left(\frac{x_{i+1} - x_i}{\Delta t} \right)^2 \Delta t \right].$$

We want to compute

$$\langle x_{i+1} | e^{-i\Delta t \frac{\hat{p}^2}{2m}} | x_i \rangle.$$

Insert a resolution of identity in momentum space,

$$\mathbf{1} = \int \frac{dp}{2\pi} |p\rangle \langle p|,$$

together with the plane-wave overlap (our convention)

$$\langle x | p \rangle = e^{ipx}, \quad \langle p | x \rangle = e^{-ipx}.$$

Then

$$\begin{aligned} \langle x_{i+1} | e^{-i\Delta t \frac{\hat{p}^2}{2m}} | x_i \rangle &= \int \frac{dp}{2\pi} \langle x_{i+1} | p \rangle \langle p | e^{-i\Delta t \frac{\hat{p}^2}{2m}} | x_i \rangle \\ &= \int \frac{dp}{2\pi} \langle x_{i+1} | p \rangle e^{-i\Delta t \frac{p^2}{2m}} \langle p | x_i \rangle \\ &= \int \frac{dp}{2\pi} \exp \left[ip(x_{i+1} - x_i) - i\Delta t \frac{p^2}{2m} \right]. \end{aligned} \quad (6)$$

Let $\Delta x \equiv x_{i+1} - x_i$. Complete the square in the exponent:

$$\begin{aligned} ip\Delta x - i\Delta t \frac{p^2}{2m} &= -\frac{i\Delta t}{2m} \left[p^2 - \frac{2m}{\Delta t} p \Delta x \right] \\ &= -\frac{i\Delta t}{2m} \left[\left(p - \frac{m\Delta x}{\Delta t} \right)^2 - \left(\frac{m\Delta x}{\Delta t} \right)^2 \right] \\ &= -\frac{i\Delta t}{2m} \left(p - \frac{m\Delta x}{\Delta t} \right)^2 + \frac{im}{2} \frac{(\Delta x)^2}{\Delta t}. \end{aligned} \quad (7)$$

Substitute this into (6):

$$\langle x_{i+1} | e^{-i\Delta t \frac{\hat{p}^2}{2m}} | x_i \rangle = \exp\left(\frac{im}{2} \frac{(\Delta x)^2}{\Delta t}\right) \int \frac{dp}{2\pi} \exp\left[-\frac{i\Delta t}{2m} \left(p - \frac{m\Delta x}{\Delta t}\right)^2\right]. \quad (8)$$

Now shift the integration variable

$$q \equiv p - \frac{m\Delta x}{\Delta t}, \quad dp = dq,$$

and use the standard Gaussian integral (with $\Delta t > 0$ understood together with the usual $i\epsilon$ prescription to define the oscillatory integral):

$$\int_{-\infty}^{\infty} \frac{dq}{2\pi} \exp\left(-i \frac{\Delta t}{2m} q^2\right) = \sqrt{\frac{m}{2\pi i \Delta t}}.$$

Therefore,

$$\begin{aligned} \langle x_{i+1} | e^{-i\Delta t \frac{\hat{p}^2}{2m}} | x_i \rangle &= \sqrt{\frac{m}{2\pi i \Delta t}} \exp\left(\frac{im}{2} \frac{(\Delta x)^2}{\Delta t}\right) \\ &= \sqrt{\frac{m}{2\pi i \Delta t}} \exp\left[\frac{im}{2} \left(\frac{x_{i+1} - x_i}{\Delta t}\right)^2 \Delta t\right], \end{aligned} \quad (9)$$

which is exactly the desired result.

- (c) Combine this with $\langle x_{i+1} | e^{-i\Delta t V(\hat{x})} | x_i \rangle = e^{-i\Delta t V(x_i)} \delta(x_{i+1} - x_i)$ to show that the full short-time propagator is

$$\langle x_{i+1} | e^{-i\hat{H}\Delta t} | x_i \rangle = \sqrt{\frac{m}{2\pi i \Delta t}} \exp\left[i\Delta t \left(\frac{1}{2} m \dot{x}_i^2 - V(x_i)\right)\right],$$

where $\dot{x}_i \equiv (x_{i+1} - x_i)/\Delta t$.

Using the BCH result from part (a), for sufficiently small Δt we may write

$$e^{-i\hat{H}\Delta t} = e^{-i\Delta t \frac{\hat{p}^2}{2m}} e^{-i\Delta t V(\hat{x})} + \mathcal{O}(\Delta t^2).$$

Insert a complete set of position eigenstates between the two factors:

$$\langle x_{i+1} | e^{-i\hat{H}\Delta t} | x_i \rangle = \int dx \langle x_{i+1} | e^{-i\Delta t \frac{\hat{p}^2}{2m}} | x \rangle \langle x | e^{-i\Delta t V(\hat{x})} | x_i \rangle + \mathcal{O}(\Delta t^2). \quad (10)$$

Since $V(\hat{x})$ is diagonal in the $|x\rangle$ basis, we have

$$\langle x | e^{-i\Delta t V(\hat{x})} | x_i \rangle = e^{-i\Delta t V(x_i)} \delta(x - x_i).$$

Substitute this into (10) and perform the x -integral:

$$\langle x_{i+1} | e^{-i\hat{H}\Delta t} | x_i \rangle = e^{-i\Delta t V(x_i)} \langle x_{i+1} | e^{-i\Delta t \frac{\hat{p}^2}{2m}} | x_i \rangle + \mathcal{O}(\Delta t^2). \quad (11)$$

Now use the free-particle matrix element from part (b):

$$\langle x_{i+1} | e^{-i\Delta t \frac{\hat{p}^2}{2m}} | x_i \rangle = \sqrt{\frac{m}{2\pi i \Delta t}} \exp \left[\frac{im}{2} \frac{(x_{i+1} - x_i)^2}{\Delta t} \right].$$

Therefore,

$$\langle x_{i+1} | e^{-i\hat{H}\Delta t} | x_i \rangle = \sqrt{\frac{m}{2\pi i \Delta t}} \exp \left[\frac{im}{2} \frac{(x_{i+1} - x_i)^2}{\Delta t} - i\Delta t V(x_i) \right] + \mathcal{O}(\Delta t^2). \quad (12)$$

Finally define the discrete velocity

$$\dot{x}_i \equiv \frac{x_{i+1} - x_i}{\Delta t},$$

so that

$$\frac{m}{2} \frac{(x_{i+1} - x_i)^2}{\Delta t} = \Delta t \frac{1}{2} m \dot{x}_i^2.$$

Hence the short-time propagator becomes

$$\langle x_{i+1} | e^{-i\hat{H}\Delta t} | x_i \rangle = \sqrt{\frac{m}{2\pi i \Delta t}} \exp \left[i\Delta t \left(\frac{1}{2} m \dot{x}_i^2 - V(x_i) \right) \right] + \mathcal{O}(\Delta t^2). \quad (13)$$

Problem 2. Path Integral and Vacuum Projection (15 points)

In Lecture 14 we used the “ $i\epsilon$ trick” to project onto the ground state in the limit $t \rightarrow +\infty$, $t' \rightarrow -\infty$.

- (a) Let $\{|n\rangle\}$ be the energy eigenbasis of a Hamiltonian H . Show that inserting a factor

$$e^{-iH(1-i\epsilon)t}$$

suppresses the contribution of all excited states as $t \rightarrow +\infty$, and derive explicitly that

$$\lim_{t \rightarrow +\infty} \langle x, t | n \rangle \propto e^{-E_n \epsilon t}.$$

Explain why only the ground state survives.

Let $\{|n\rangle\}$ be a complete set of energy eigenstates,

$$H|n\rangle = E_n|n\rangle, \quad E_0 \leq E_1 \leq E_2 \leq \dots.$$

In the Heisenberg time-evolution operator we insert a small imaginary part in time (or equivalently in the Hamiltonian),

$$U_\epsilon(t) \equiv e^{-iH(1-i\epsilon)t}, \quad \epsilon > 0.$$

First expand the exponential:

$$-iH(1-i\epsilon)t = -iHt - \epsilon Ht,$$

so

$$U_\epsilon(t) = e^{-iHt} e^{-\epsilon Ht}, \tag{14}$$

where $e^{-\epsilon Ht}$ is *non-unitary* and will damp higher-energy components. Now consider the position-time bra

$$\langle x, t | \equiv \langle x | U_\epsilon(t) = \langle x | e^{-iH(1-i\epsilon)t}.$$

Insert a complete set of energy eigenstates:

$$\begin{aligned} \langle x, t | n \rangle &= \langle x | e^{-iH(1-i\epsilon)t} | n \rangle \\ &= \langle x | e^{-iE_n(1-i\epsilon)t} | n \rangle \quad (\text{since } H|n\rangle = E_n|n\rangle) \\ &= e^{-iE_n t} e^{-E_n \epsilon t} \langle x | n \rangle. \end{aligned} \tag{15}$$

Therefore,

$$\langle x, t|n\rangle = e^{-iE_n t} e^{-E_n \epsilon t} \psi_n(x), \quad \psi_n(x) \equiv \langle x|n\rangle, \quad (16)$$

and in particular

$$\lim_{t \rightarrow +\infty} \langle x, t|n\rangle \propto e^{-E_n \epsilon t}. \quad (17)$$

Why only the ground state survives. The factor $e^{-E_n \epsilon t}$ exponentially suppresses larger energies when $t \rightarrow +\infty$ (with $\epsilon > 0$ fixed):

$$\frac{\langle x, t|n\rangle}{\langle x, t|0\rangle} = e^{-i(E_n - E_0)t} e^{-(E_n - E_0)\epsilon t} \frac{\psi_n(x)}{\psi_0(x)} \xrightarrow{t \rightarrow +\infty} 0 \quad (n > 0),$$

because $E_n - E_0 > 0$ for any excited state. Equivalently, for a generic state $|\Psi\rangle = \sum_n c_n |n\rangle$,

$$e^{-iH(1-i\epsilon)t} |\Psi\rangle = \sum_n c_n e^{-iE_n t} e^{-E_n \epsilon t} |n\rangle = e^{-iE_0 t} e^{-E_0 \epsilon t} \left[c_0 |0\rangle + \sum_{n>0} c_n e^{-(E_n - E_0)\epsilon t} e^{-i(E_n - E_0)t} |n\rangle \right],$$

and the bracket approaches $c_0 |0\rangle$ as $t \rightarrow +\infty$. Thus, provided the initial state has nonzero overlap with the ground state ($c_0 \neq 0$), the $i\epsilon$ deformation projects onto $|0\rangle$.

(b) Using the result of part (a), show that for an operator X

$$\lim_{t \rightarrow \infty, t' \rightarrow -\infty} \langle x, t|X|x', t'\rangle = \psi_0(x) \psi_0^*(x') e^{-iE_0(t-t')} \langle 0|X|0\rangle,$$

where $\psi_0(x) = \langle x|0\rangle$.

Use the $i\epsilon$ -deformed time evolution on both sides:

$$\langle x, t|X|x', t'\rangle \equiv \langle x|e^{-iH(1-i\epsilon)t} X e^{+iH(1-i\epsilon)t'}|x'\rangle, \quad (\epsilon > 0).$$

Insert complete sets of energy eigenstates $\mathbf{1} = \sum_n |n\rangle \langle n|$:

$$\langle x, t|X|x', t'\rangle = \sum_{n,m} \langle x|e^{-iH(1-i\epsilon)t}|n\rangle \langle n|X|m\rangle \langle m|e^{+iH(1-i\epsilon)t'}|x'\rangle. \quad (18)$$

Now evaluate the time-evolution factors using $H|n\rangle = E_n|n\rangle$. From part (a),

$$\langle x|e^{-iH(1-i\epsilon)t}|n\rangle = e^{-iE_n t} e^{-E_n \epsilon t} \psi_n(x), \quad \psi_n(x) \equiv \langle x|n\rangle.$$

Similarly,

$$\begin{aligned}
 \langle m | e^{+iH(1-i\epsilon)t'} | x' \rangle &= \left(\langle x' | e^{-iH(1-i\epsilon)t'} | m \rangle \right)^* \\
 &= \left(e^{-iE_m t'} e^{-E_m \epsilon t'} \psi_m(x') \right)^* \\
 &= e^{+iE_m t'} e^{-E_m \epsilon |t'|} \psi_m^*(x'), \quad (t' \rightarrow -\infty \Rightarrow |t'| = -t'). \quad (19)
 \end{aligned}$$

Substituting into (18) gives

$$\langle x, t | X | x', t' \rangle = \sum_{n,m} \psi_n(x) \psi_m^*(x') e^{-iE_n t} e^{+iE_m t'} e^{-E_n \epsilon t} e^{-E_m \epsilon |t'|} \langle n | X | m \rangle. \quad (20)$$

Take the limit $t \rightarrow +\infty$ and $t' \rightarrow -\infty$. Because $E_n \geq E_0$ and $\epsilon > 0$, the damping factors $e^{-E_n \epsilon t}$ and $e^{-E_m \epsilon |t'|}$ suppress all terms with $n > 0$ or $m > 0$ relative to the $(0,0)$ term. More explicitly,

$$\frac{\text{term}(n, m)}{\text{term}(0, 0)} \sim e^{-(E_n - E_0) \epsilon t} e^{-(E_m - E_0) \epsilon |t'|} \xrightarrow{t \rightarrow \infty, t' \rightarrow -\infty} 0 \quad \text{if } (n, m) \neq (0, 0).$$

Therefore the double sum is dominated by $(n, m) = (0, 0)$:

$$\begin{aligned}
 \lim_{t \rightarrow \infty, t' \rightarrow -\infty} \langle x, t | X | x', t' \rangle &= \psi_0(x) \psi_0^*(x') e^{-iE_0 t} e^{+iE_0 t'} \langle 0 | X | 0 \rangle \\
 &= \boxed{\psi_0(x) \psi_0^*(x') e^{-iE_0(t-t')} \langle 0 | X | 0 \rangle}. \quad (21)
 \end{aligned}$$

This is the desired result.

(c) Explain why taking the ratio

$$\frac{\langle x, t | X | x', t' \rangle}{\langle x, t | x', t' \rangle} \quad \text{as } t \rightarrow +\infty, t' \rightarrow -\infty$$

removes the wavefunctions and the divergent normalization constants, yielding

$$\langle 0 | X | 0 \rangle.$$

From part (b), for any operator X we have, in the limits $t \rightarrow +\infty$ and $t' \rightarrow -\infty$,

$$\langle x, t | X | x', t' \rangle \longrightarrow \psi_0(x) \psi_0^*(x') e^{-iE_0(t-t')} \langle 0 | X | 0 \rangle. \quad (22)$$

Apply the same reasoning to the denominator with $X = \mathbf{1}$:

$$\langle x, t | x', t' \rangle = \langle x, t | \mathbf{1} | x', t' \rangle \longrightarrow \psi_0(x) \psi_0^*(x') e^{-iE_0(t-t')}. \quad (23)$$

Here the matrix element $\langle 0|1|0\rangle = 1$, so the denominator contains only the ground-state wavefunctions and the same time-dependent phase factor. Taking the ratio of (22) and (23), we find

$$\begin{aligned} \frac{\langle x, t|X|x', t'\rangle}{\langle x, t|x', t'\rangle} &\xrightarrow{t \rightarrow \infty, t' \rightarrow -\infty} \frac{\psi_0(x)\psi_0^*(x') e^{-iE_0(t-t')}}{\psi_0(x)\psi_0^*(x') e^{-iE_0(t-t')}} \frac{\langle 0|X|0\rangle}{1} \\ &= \boxed{\langle 0|X|0\rangle}. \end{aligned} \quad (24)$$

Physical interpretation.

- The factors $\psi_0(x)\psi_0^*(x')$ depend on the arbitrary external position states $|x\rangle$ and $|x'\rangle$ and therefore carry no intrinsic physical information about the operator X .
- The oscillatory phase $e^{-iE_0(t-t')}$ and any divergent normalization associated with the vacuum propagation appear in both numerator and denominator and cancel exactly.
- The ratio thus removes all kinematic and normalization factors and projects onto the intrinsic vacuum expectation value of X .

This is the conceptual reason why, in path-integral formulations, vacuum expectation values are obtained by dividing correlation functions by the vacuum persistence amplitude.

- (d) Apply this to the operator insertion $X = T(x(t_1)x(t_2))$ and write the path-integral expression for the two-point function

$$G_2(t_1, t_2) = \langle 0|T(x(t_1)x(t_2))|0\rangle.$$

Take the operator insertion

$$X = T(\hat{x}(t_1)\hat{x}(t_2)).$$

From part (c), the vacuum expectation value is obtained by the ratio

$$\langle 0|X|0\rangle = \lim_{t \rightarrow +\infty, t' \rightarrow -\infty} \frac{\langle x, t|X|x', t'\rangle}{\langle x, t|x', t'\rangle}.$$

Using the time-sliced path-integral representation of the transition amplitude (with the usual $i\epsilon$ prescription understood to project onto the ground state), the numer-

ator becomes a path integral with the insertion $x(t_1)x(t_2)$:

$$\langle x, t | T(\hat{x}(t_1)\hat{x}(t_2)) | x', t' \rangle = \int_{x(t')=x'}^{x(t)=x} \mathcal{D}x(\tau) x(t_1) x(t_2) \exp[iS_\epsilon[x]], \quad (25)$$

and the denominator is the same path integral without insertion:

$$\langle x, t | x', t' \rangle = \int_{x(t')=x'}^{x(t)=x} \mathcal{D}x(\tau) \exp[iS_\epsilon[x]]. \quad (26)$$

Therefore the two-point function can be written as

$$\begin{aligned} G_2(t_1, t_2) &\equiv \langle 0 | T(\hat{x}(t_1)\hat{x}(t_2)) | 0 \rangle \\ &= \lim_{t \rightarrow +\infty, t' \rightarrow -\infty} \frac{\int_{x(t')=x'}^{x(t)=x} \mathcal{D}x(\tau) x(t_1) x(t_2) e^{iS_\epsilon[x]}}{\int_{x(t')=x'}^{x(t)=x} \mathcal{D}x(\tau) e^{iS_\epsilon[x]}}. \end{aligned} \quad (27)$$

In practice one usually writes this directly as a vacuum functional integral over histories defined on the whole line $\tau \in (-\infty, +\infty)$ (with the $i\epsilon$ prescription enforcing vacuum boundary conditions):

$$G_2(t_1, t_2) = \frac{\int \mathcal{D}x(\tau) x(t_1) x(t_2) e^{iS_\epsilon[x]}}{\int \mathcal{D}x(\tau) e^{iS_\epsilon[x]}}. \quad (28)$$

Here $S_\epsilon[x]$ is the action with the small imaginary part (the $i\epsilon$ trick) that projects onto the ground state.

Problem 3. Bonus (optional): From QM to QFT (5 points)

Explain why the replacement

$$x(t) \longrightarrow \phi(t, \mathbf{x}), \quad \int dt \longrightarrow \int d^4x,$$

together with the identification

$$S[\phi] = \int d^4x \left(-\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi) \right),$$

leads directly from the QM path integral to the QFT functional integral

$$Z = \int \mathcal{D}\phi e^{iS[\phi]}.$$

Which conceptual step from Lecture 14 guarantees that fields at different spatial points act like infinitely many decoupled degrees of freedom in the free theory?

In non-relativistic quantum mechanics, the basic dynamical variable is a single coordinate $x(t)$, and the path integral sums over all possible *time histories* of this coordinate:

$$\int \mathcal{D}x(t) \exp \left[i \int dt \left(\frac{1}{2} m \dot{x}^2 - V(x) \right) \right].$$

In quantum field theory, the dynamical variable is no longer a single degree of freedom, but a field $\phi(t, \mathbf{x})$ defined at every point in space. Conceptually, this means that at each spatial point $\mathbf{x}$ we have an independent coordinate $\phi_{\mathbf{x}}(t)$. Thus the replacement

$$x(t) \longrightarrow \phi(t, \mathbf{x})$$

turns the single path $x(t)$ into an infinite collection of paths, one for each spatial point. Correspondingly, the time integral becomes a spacetime integral,

$$\int dt \longrightarrow \int d^4x,$$

and the action generalizes to

$$S[\phi] = \int d^4x \left(-\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi) \right).$$

Summing over all possible field configurations then leads directly to the functional integral

$$Z = \int \mathcal{D}\phi e^{iS[\phi]}.$$

The crucial idea introduced in Lecture 14 is that the *free-field Hamiltonian is quadratic*. After Fourier transforming in space,

$$\phi(t, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \phi_{\mathbf{k}}(t) e^{i\mathbf{k}\cdot\mathbf{x}},$$

the free action becomes a sum of independent harmonic oscillators, one for each momentum mode $\mathbf{k}$:

$$S_0 = \int dt \sum_{\mathbf{k}} \left[\frac{1}{2} |\dot{\phi}_{\mathbf{k}}|^2 - \frac{1}{2} \omega_{\mathbf{k}}^2 |\phi_{\mathbf{k}}|^2 \right].$$

Because the action is a *sum of decoupled quadratic terms*, the path integral factorizes into infinitely many independent Gaussian integrals. This guarantees that, in the free theory, field degrees of freedom at different spatial points (or equivalently different momentum modes) behave like infinitely many decoupled quantum-mechanical oscillators.

Summary.

- QM describes one degree of freedom $x(t)$; QFT describes infinitely many degrees of freedom $\phi(t, \mathbf{x})$.
- The path integral generalizes by summing over all field configurations instead of paths.
- The quadratic (free) action ensures decoupling, making the QFT path integral a direct generalization of the QM one.